

EFFECTIVE TRACE BOUNDS AND RIGIDITY FOR SIMPLE CURVES ON THE MODULAR TORUS

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ABSTRACT. We study equal traces of primitive simple curves on the modular once-punctured torus, with the relative occurrence count fixed and the integral Markov character allowed to vary. For every occurrence count we give a complete decision procedure, polynomial in that count, together with an exact test for surface isometry. The proof combines positive matrix paths with a controlled comparison between actual Markov triples and commuting trace triples: positive quotient mutation bounds the error at positive product index, while at minimum index mechanical balance and spectral coordinates control its propagation along the Euclidean algorithm. Both parameter bounds are analytic, valid over all positive integral characters, and are stated explicitly in the text. An exact finite computation classifies all primitive occurrence families with between two and six occurrences; every equal pair in these families is isometric. Two further arguments give the same conclusion for every pair of intersection number at most twenty-four, so that a nonisometric equality would need intersection at least twenty-five. These results leave the occurrence count unbounded and do not resolve Markov uniqueness.

1. INTRODUCTION

A simple closed geodesic on the modular punctured torus carries an integer: its hyperbolic trace divided by three. The integers obtained in this way are the Markov numbers. Their construction remembers a curve, a rational slope, or a position in a tree of Markov triples. The number itself forgets this information. The Markov–Frobenius uniqueness conjecture [7] asks whether that loss of information creates any ambiguity beyond the symmetries already present in the construction. In its geometric form, it asks whether two simple closed geodesics of equal length on the modular torus must be related by an isometry.

This paper studies the question with a parameter attached to the *pair* of curves. Choose one curve as the first element A of a simple basis (A, B) . The second curve then has primitive homology (h, N) . For a normalized representative $2 \leq h < N$, write $N = hk + t$ with $k \geq 1$ and $0 < t < h$, and a Christoffel word represents it as h occurrences of A separated by gaps B^k and B^{k+1} . We fix the occurrence count h and allow both the gap quotient k and the integral character of the basis to vary. The question is whether the trace of this word can equal the trace of A . The endpoint cases $N = 1$ and $h = 1$ are covered separately by a reflection argument.

There are two obstacles to making this a finite problem. First, a basis with very large traces can still represent a curve of small trace when it is far from the minimum of its Markov ray. Second, equality of traces must be separated from equality explained by an isometry of the surface. An arbitrary change of homology basis answers neither question. Our argument centers each ray at its actual integral minimum, bounds all remaining indices and parameters, and retains the marking throughout the resulting finite calculation.

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1.1. Main conclusions. Our main quantitative result is Theorem 6.3. For each $h \geq 2$, every primitive mechanical trace equality in the preceding family lies in a fully explicit domain with $O(h^5)$ exact comparisons on $O(h^3)$ -bit integers. The isometry status of every equality is decided by twelve fixed integral actions. This is a polynomial bound in the numerical value of h , rather than in the length of its binary encoding.

The two parameter estimates behind this result are particularly simple. Write $p = \text{tr } B/3$ after centering the ray. If the base index l of the product is positive, then

$$p < 4^{h+1}.$$

If $l = 0$, mechanical balance gives

$$p < (100h^2)^{h+1}.$$

These are analytic bounds, valid over all positive integral Markov characters. Although their heights are large, the number of Markov occurrences below them is small enough for a polynomial decision bound. Counting occurrences instead of scanning integer labels is an essential part of the argument.

The finite applications give two complete rigidity statements: all primitive occurrence families $2 \leq h \leq 6$ contain only isometric equalities, and every equal-length simple pair with intersection number at most 24 is isometric. No upper bound on the common trace is assumed. Both proofs combine the analytic reductions with an exact finite computation, together with separate arguments for the remaining intersection-19 and intersection-23 types; the computation is described where it is used, and the data it produces accompany the article as supplementary material. Section 7 specifies the geometric normalization and the precise scope of these conclusions.

1.2. Why a comparison with commuting traces helps. The mechanism is an integrality contradiction. If $s > 1$, the commuting baselines

$$B_i = \frac{s^i + s^{-i}}{3}$$

satisfy $3B_a B_b - B_{a+b} = B_{|a-b|}$. Their indices therefore follow the Euclidean algorithm under Vieta mutation. When the indices reach zero, the corresponding baseline is $B_0 = 2/3$. Actual positive integral Markov triples obey the same polynomial mutation, but lie on a different level of the trace cubic. If their errors relative to the baselines remain sufficiently small, the final actual entry would be a positive integer below one.

The comparison has a nonzero forcing term, so small initial errors alone do not suffice. At positive product index, the quotient form of Vieta mutation keeps a weighted error bounded by the same constant at every step. At minimum index, cyclic balance first bounds the mixed matrix paths. The equality then improves the relevant block exponents to positive multiples of a greatest common divisor. Passing to spectral coordinates turns a mutation into subtraction with a small additive error, whose propagation is controlled by the integer rows of the Euclidean algorithm. This is the reason for using two complementary error estimates.

1.3. Relation to earlier work. The correspondence between Markov arithmetic and the geometry of the punctured torus is classical. Cohn related Markov forms to simple geodesics [5], and Reutenauer gave a constructive bijection between proper Christoffel words and proper Markov triples [13, Theorem 1]. These correspondences retain a whole triple or a marked curve. They do not assert that its largest coordinate, or the trace of the curve alone, determines it. We use the classical dictionaries to enumerate marked occurrences and retain their homology throughout the calculation.

The parameter in this paper is attached to a pair of curves. After choosing one member as a basis element, the other has primitive homology $(h, hk+t)$ and a prescribed Christoffel

gap pattern. This differs from fixing a numerator in one global Farey parametrization. The ordering theorems of Lagisquet, Pelantová, Tavenas and Vuillon [9] and of Lee, Li, Rabideau and Schiffler [10] compare Markov values along specified directions in that global parametrization. Our bounds allow the integral character of the chosen basis to vary. Changing the basis transforms both curves and their character; it cannot be used to reset the first curve while leaving its trace equation unchanged.

Effective finiteness on character varieties is also established. Whang proves effective bounds for integral points on geometrically irreducible nonintegrable curves in relative character varieties, and describes the multitwist exceptions in the more general nondegenerate case [14, Theorems 1.2–1.3]. Applied to a fixed trace-equality curve, this method requires verification of the exceptional hypotheses component by component. The present argument gives explicit bounds in the pair parameter h and a direct finite comparison domain, without computing those algebraic curves. Its quantitative conclusion is $O(h^5)$ trace comparisons on $O(h^3)$ -bit integers. This is a bound in the numerical value of h , and does not bound h itself.

The geometric part uses equally classical structure. The length of a primitive simple class on a punctured torus is determined by its primitive homology vector, and these lengths extend to a norm [12]. McShane and Parlier prove that, with the boundary length fixed, the locus where two distinct simple curves have equal length is an embedded path [11, Theorem 1.3]. We use this theorem for the reflection cases. The remaining arithmetic classifications retain the full marking and test the actual finite isometry group of the modular torus. Homological equivalence under an arbitrary change of basis is not an isometry test at a fixed metric.

The comparison model for the analytic estimates has a useful interpretation on the Cayley cubic. For full traces, put

$$F(X, Y, Z) = X^2 + Y^2 + Z^2 - XYZ.$$

The Fricke identity gives $F = \text{tr}[A, B] + 2$, so the cusped Markov character lies on $F = 0$ [8]. Commuting diagonal matrices instead give

$$(X, Y, Z) = (u + u^{-1}, v + v^{-1}, uv + (uv)^{-1}), \quad F = 4.$$

The torus cover and monomial action on this second level are classical [4, Sections 1.5 and 2.1]. Taking $(u, v) = (s^a, s^b)$ turns a Vieta move into subtraction of the exponents. The same polynomial move preserves the actual Markov level separately. Our estimates control the discrepancy along these two evolutions; after division of the traces by three, their level difference is $4/9$. There is no identification of the two character surfaces in this argument.

Positive trace expansions and quadratic norm gaps have substantial prior uses. Fisac’s necklace formula already records the dependence of a trace on cyclic runs [6, Theorem 1.1 and Section 4]. Altassan and Luca use Binet expansions and quadratic norms to bound solutions of Markov-type equations in Lucas sequences [1, Theorem 1.1 and Lemmas 2.1–2.2]. Their sequences have specified initial values. Here a centered Markov ray has coefficients satisfying $cd = p^2/(9p^2 - 4)$, and these coefficients vary with the integral character. The needed estimates therefore retain both coefficients, all mixed matrix paths and the actual Vieta mutations. At minimum index, mechanical balance controls the mixed paths, while spectral-angle errors propagate through the Euclidean algorithm with polynomial amplification. The contradiction is an actual positive integer forced below one.

Finally, arithmetic restrictions on the largest Markov entry, such as Button’s ideal-theoretic criteria [3], organize partial uniqueness by a different parameter. The applications here concern complete families of relative curve positions: occurrence counts two

through six, and all intersections at most twenty-four. No upper bound on the common Markov value is assumed in these classifications. These classifications and the effective decision procedure leave open whether a nonisometric equality occurs at some larger occurrence count.

1.4. Organization. Section 2 fixes the curve, word and ray conventions. Section 3 bounds the product index and the original gap quotient using positive paths and a quadratic norm. Sections 4 and 5 prove the two parameter bounds. Section 6 gives the complete finite domain and its isometry test. Section 7 applies the procedure to the stated families; the exceptional intersection argument and the description of the finite domains are in the appendix. The final section explains which part of the uniqueness problem remains.

2. SIMPLE CURVES, MECHANICAL WORDS, AND POSITIVE MARKOV RAYS

2.1. The geometric normalization. We use the complete hyperbolic metric of curvature -1 on the modular once-punctured torus T . A convenient holonomy basis is

$$E = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad F = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}. \quad (1)$$

Its commutator has trace -2 . The image of $\langle E, F \rangle$ in $\mathrm{PSL}_2(\mathbb{R})$ uniformizes T . An essential simple closed curve means a nonperipheral, unoriented simple free homotopy class; its geodesic representative has length

$$\ell(\gamma) = 2 \operatorname{arccosh}(|\operatorname{tr} \gamma|/2). \quad (2)$$

Thus equal length is precisely equal absolute trace. We choose positive lifts for the simple generators occurring below.

The classical Markov dictionary identifies positive normalized traces of a simple basis and its product with positive integral solutions of

$$a^2 + b^2 + c^2 = 3abc. \quad (3)$$

Here and throughout, a normalized trace is the ordinary matrix trace divided by three. Changing one vertex of a Farey triangle replaces one coordinate, say c , by $3ab - c$. This follows directly from the SL_2 identity

$$\operatorname{tr}(XY) + \operatorname{tr}(XY^{-1}) = \operatorname{tr} X \operatorname{tr} Y.$$

The initial traces in (1) give $(1, 1, 1)$. Conversely, largest-coordinate descent generates all positive integral solutions of (3); Section 6 makes this construction explicit. We use the classical geometric identification of these traces with simple curves, as developed by Cohn and Goldman [5, 8]. Equal numerical labels at different occurrences of this construction are never identified in our algorithm.

2.2. Which words represent simple curves? Fix oriented simple basis curves (A, B) meeting once, with homology columns $\alpha, \beta \in H_1(T; \mathbb{Z}) \simeq \mathbb{Z}^2$. Essential simple classes are in bijection with primitive vectors, up to sign. Their geometric intersection number is

$$i(v, w) = |\det(v, w)|. \quad (4)$$

In particular, a pair of distinct classes can be written in a common integral basis as $(1, 0)$ and (h, N) , where $N > 0$, $0 \leq h < N$, and $\gcd(h, N) = 1$. Changing the complementary basis vector changes h by a multiple of N . Reversing orientations and interchanging the two curves gives the residue orbit

$$\{h, -h, h^{-1}, -h^{-1}\} \pmod{N}. \quad (5)$$

This is a change of marking of the pair. It does not assert that these changes of marking are isometries of the fixed surface.

Suppose $2 \leq h < N$, and write $N = hk + t$, with $k \geq 1$, $0 < t < h$, and $\gcd(h, t) = 1$. The primitive simple class of slope (h, N) is represented, up to cyclic conjugacy, by the mechanical word

$$W_{h,k,t}(A, B) = \prod_{i=1}^h AB^{k+\epsilon_i}, \quad \epsilon_i = \left\lfloor \frac{it}{h} \right\rfloor - \left\lfloor \frac{(i-1)t}{h} \right\rfloor. \quad (6)$$

Its h gaps contain exactly t ones. This is the Christoffel cutting-sequence description of a primitive straight line on a torus; see [13]. One can see the formula by counting the horizontal grid lines crossed in each successive vertical strip. Coprimality makes the sequence a single primitive period. Translating the starting point changes only the cyclic conjugacy class. Inverting one coordinate reverses and complements the corresponding gap pattern.

We call h the *occurrence count* in this marking. The mechanical condition matters: an arbitrary word with primitive abelianization need not be a simple curve. Some analytic estimates below hold for every binary pattern, but only mechanical patterns enter the simple-curve decision procedure.

2.3. Centering a Markov ray at its minimum. Keep B fixed and replace A by AB^j , for $j \in \mathbb{Z}$. If $\text{tr } B = 3p$, put $u_j = \text{tr}(AB^j)/3$. The trace identity and (3) give

$$u_{j+1} = 3pu_j - u_{j-1}, \quad p^2 + u_j^2 + u_{j+1}^2 = 3pu_ju_{j+1}. \quad (7)$$

These are positive integers at every integral index: each adjacent triple is a positive Markov triple, and its Vieta replacement is $(p^2 + u_j^2)/u_{j+1} > 0$. We call this a positive integral Markov ray. The following elementary normalization will be used repeatedly.

Lemma 2.1 (The positive ray model). *Let $p \geq 1$ and let $(u_j)_{j \in \mathbb{Z}}$ be a positive integral Markov ray. Define*

$$D = 9p^2 - 4, \quad \lambda = \frac{3p + \sqrt{D}}{2}, \quad K = \frac{p^2}{D}. \quad (8)$$

After shifting an index at which u_j is minimal to zero, there are positive conjugate elements $c, d \in \mathbb{Q}(\sqrt{D})$ such that

$$u_j = c\lambda^j + d\lambda^{-j}, \quad cd = K, \quad \lambda^{-1} \leq c/d \leq \lambda. \quad (9)$$

Furthermore, $\sqrt{D}c$ and $\sqrt{D}d$ are algebraic integers, and

$$\frac{5}{2}p < \lambda < 3p, \quad 5p^2 \leq D < 9p^2, \quad \frac{1}{9} < K \leq \frac{1}{5}, \quad c, d < \sqrt{p}. \quad (10)$$

A marked realization with this ray is simultaneously conjugate over $\mathbb{R}$ to

$$A_0 = \begin{pmatrix} 3c & 2/\sqrt{D} \\ 2/\sqrt{D} & 3d \end{pmatrix}, \quad B = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda^{-1} \end{pmatrix}. \quad (11)$$

Proof. The integer D lies strictly between $(3p-1)^2$ and $(3p)^2$, so it is not a square. The characteristic roots of the recurrence are λ and λ^{-1} , and

$$c = \frac{u_1 - \lambda^{-1}u_0}{\sqrt{D}}, \quad d = \frac{\lambda u_0 - u_1}{\sqrt{D}}.$$

Conjugation sends λ to λ^{-1} and $\sqrt{D}$ to $-\sqrt{D}$, so it exchanges c, d . Substituting the Markov equation gives $cd = p^2/D > 0$. Since $c + d = u_0 > 0$, both are positive. The expressions also show that $\sqrt{D}c$ and $\sqrt{D}d$ are algebraic integers: $\lambda^{\pm 1}$ are roots of the monic polynomial $X^2 - 3pX + 1$.

The ray tends to infinity at both ends and therefore has an integral minimum. The inequalities $u_0 \leq u_1, u_{-1}$ are exactly $c/d \geq \lambda^{-1}$ and $c/d \leq \lambda$. The bounds for D, K, λ follow from their definitions; in particular $\sqrt{9p^2 - 4} > 2p$. Hence $c^2, d^2 \leq K\lambda < 3p/5 < p$.

Finally diagonalize the hyperbolic matrix B . The traces of A_0 and A_0B determine its diagonal entries to be $3c, 3d$. Its unit determinant forces the off-diagonal product to equal $9cd - 1 = 4/D > 0$. A real diagonal conjugation, including a sign change if necessary, makes both off-diagonal entries $2/\sqrt{D}$. $\square$

Lemma 2.2 (The integral minimum bound). *If $x = u_0$ is a minimum of a positive integral Markov ray, then*

$$(3p - 2)x^2 \leq p^2. \quad (12)$$

For $p \geq 3$ this strengthens to $3x^2 \leq p$.

Proof. The two neighbors of x are the roots of $f(T) = T^2 - 3pxT + p^2 + x^2$. Both are at least x , so $f(x) \geq 0$, which gives (12). For $p \geq 2$,

$$3x^2 \leq \frac{3p^2}{3p-2} = p + \frac{2p}{3p-2} \leq p + 1.$$

If $3x^2 > p$, integrality forces $3x^2 = p + 1$. Substituting $p = 3x^2 - 1$ into (12), $x^2 \leq 1$. It would therefore force $p = 2, x = 1$, where $3x^2 = 3 > 2$ actually occurs. Thus the stronger bound holds for $p \geq 3$; the exceptional minimum $(p, x) = (2, 1)$ must be retained separately. $\square$

At $p = 1, 2$ the minimum is $x = 1$. These two initial cases will be included explicitly in the finite decision domain.

3. POSITIVE TRACE PATHS AND A UNIFORM GAP BOUND

The first task is to bound the distance, along a Markov ray, between the single curve and the factors of the comparison word. Positivity controls the contribution of all matrix paths except one. An algebraic norm then prevents the coefficient of that path from agreeing exactly with the leading coefficient of the single curve. The resulting bound depends only on the number of occurrences of the first generator.

Theorem 3.1 (The gap bound). *Let (A, B) be a marked modular-torus basis with positive integral normalized Markov traces. Let $h \geq 2, k \geq 1$, and let $\epsilon_1, \dots, \epsilon_h \in \{0, 1\}$ have sum t , where $1 \leq t < h$. If*

$$\text{tr } A = \text{tr } \prod_{i=1}^h AB^{k+\epsilon_i}, \quad (13)$$

then

$$k \leq b(h) := 3h^2 + 3h - 2 + \left\lfloor \frac{3h}{2} \right\rfloor. \quad (14)$$

No balance or coprimality hypothesis on the gap pattern is required for this trace inequality. For the primitive simple-curve application, the pattern is cyclically mechanical and $\gcd(h, t) = 1$.

3.1. The positive model and the two index orientations. We use the minimum-ray normalization from Section 2. In particular,

$$\begin{aligned} u_j &= c\lambda^j + d\lambda^{-j}, & D &= 9p^2 - 4, & \lambda &= \frac{3p + \sqrt{D}}{2}, & K &= cd = \frac{p^2}{D}, \\ \lambda^{-1} &\leq c/d \leq \lambda, & c, d &< \sqrt{p}, & \frac{5}{2}p &< \lambda < 3p, & 5p^2 &\leq D < 9p^2, & \frac{1}{9} &< K \leq \frac{1}{5}. \end{aligned} \quad (15)$$

Here p and every u_j are positive integers, and u_0 is a minimum over all integral indices. Real conjugation gives the trace model

$$A_0 = \begin{pmatrix} 3c & 2/\sqrt{D} \\ 2/\sqrt{D} & 3d \end{pmatrix}, \quad B = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda^{-1} \end{pmatrix}. \quad (16)$$

Indeed the diagonal entries of the shifted first generator are $3c, 3d$. Its off-diagonal product is $9cd - 1 = 4/D > 0$, so a diagonal conjugation makes both off-diagonal entries equal to $2/\sqrt{D}$. This is a change of matrix coordinates, not an isometry assertion. Every $A_0 B^j$ is strictly positive, even for negative j , and has determinant one. Every product used below consequently has trace greater than two.

After translating the minimum to zero, let the original single generator have index n_0 and the product have base index j_0 . Then $j_0 - n_0 = k$. If $j_0 \geq 0$, put $l = j_0$. If $j_0 < 0$, reverse the whole ray and use

$$-j_0 - \epsilon_i = (-j_0 - 1) + (1 - \epsilon_i). \quad (17)$$

Thus $l = -j_0 - 1 \geq 0$, the gap bits are complemented, and their sum changes from t to $h - t$. Both sums lie strictly between zero and h . We henceforth use t for the sum in the resulting orientation. Write the single index as ϵm , with $m \geq 0$. The original quotient is recovered by one of the two formulas

$$k = l - \epsilon m \quad \text{or} \quad k = \epsilon m - l - 1, \quad k \leq l + m. \quad (18)$$

The second formula accompanies the complementary pattern. These formulas retain both directions of the ray and every possible position of its minimum.

The single index cannot be zero in (13). Before reversal, that case would have $n_0 = 0$ and $j_0 = k \geq 1$. The constant upper-state contribution to the product trace divided by three is

$$\frac{(3c\lambda^{j_0})^h \lambda^t}{3}.$$

The minimum ratio gives $c^2 \geq K/\lambda$, and $9K > 1$, so $3c\lambda > \sqrt{\lambda}$. Since $h \geq 2$ and $t \geq 1$, this contribution alone exceeds

$$\lambda^2/3 > \frac{25}{12}p^2 > 2\sqrt{p} > c + d = u_0.$$

Every other contribution is positive. Equality is therefore impossible, and from now on $m \geq 1$.

3.2. The exact path expansion. Put

$$V_l = \frac{1}{3} \operatorname{tr} \prod_{i=1}^h A_0 B^{l+\epsilon_i}. \quad (19)$$

A term in the trace is specified by a cyclic choice of upper and lower states at the h factor boundaries. Let j be its number of lower positions, let $2r$ be its number of transitions, and let e count the long-gap positions among its lower positions. The number of transitions is even because the path closes. Its exact contribution is

$$\frac{(3c)^{h-j-r} (3d)^{j-r} (4/D)^r}{3} \lambda^{(h-2j)l+t-2e}. \quad (20)$$

The exponents $h - j - r$ and $j - r$ count diagonal steps and are nonnegative. All terms in (20) are positive. Using $cd = p^2/D$, the same coefficient can be written as

$$\frac{(3c)^{h-j} (3d)^j}{3} \sigma^r, \quad \sigma = \frac{4}{9p^2} < 1. \quad (21)$$

The two paths without transitions contribute

$$X_l = 3^{h-1} c^h \lambda^{hl+t}, \quad X_l^- = 3^{h-1} d^h \lambda^{-hl-t}. \quad (22)$$

Write $V_l = X_l + \mathcal{R}_l$, where $\mathcal{R}_l > 0$.

The relative contribution of a path is

$$\sigma^r(d/c)^j \lambda^{-2jl-2e}. \quad (23)$$

It is at most $\lambda^{j(1-2l)}$. There are $\binom{h}{j}$ indexed state paths with j lower positions, irrespective of their transition counts. Summing gives

$$1 < \frac{V_l}{X_l} \leq \begin{cases} (1 + \lambda^{-1})^h, & l \geq 1, \\ (1 + \lambda)^h, & l = 0. \end{cases} \quad (24)$$

This use of the binomial theorem is an analytic sum for arbitrary h ; it does not require listing the 2^h paths.

We shall also need an absolute remainder estimate. Every path in $\mathcal{R}_l$ has at least one lower position, so its exponent in (20) is at most $(h-2)l+t$. Every entry of A_0 is less than $3\sqrt{p}$. Hence

$$0 < \mathcal{R}_l < (2^h - 1) 3^{h-1} p^{h/2} \lambda^{(h-2)l+t}. \quad (25)$$

The constant lower-state path is included in this estimate.

Lemma 3.2 (Positivity of the leading discrepancy). *Suppose a positive word of $h \geq 2$ factors in the model (16), with any integral diagonal gaps, has normalized trace $u_{\varepsilon m}$, where $m \geq 1$. Let X denote either of its constant-state contributions. If $Y = a\lambda^m$, with $a = c$ for $\varepsilon = 1$ and $a = d$ for $\varepsilon = -1$, then $Y > X$.*

Proof. The two constant contributions have product $C = 3^{2h-2} K^h = K(9K)^{h-1} > K$. The normalized word trace therefore exceeds $X + C/X$. On the other hand the single trace is $Y + K/Y$, and the minimum ratio gives $Y^2/K \geq \lambda^{2m-1} > 1$. If $Y \leq X$, monotonicity of $z + K/z$ on $z > \sqrt{K}$ yields

$$V > X + C/X > X + K/X \geq Y + K/Y,$$

a contradiction. The argument applies separately to either constant path; it does not require that the chosen one dominate the other. $\square$

3.3. A finite window for the single exponent. At an equality $V_l = u_{\varepsilon m}$, write

$$(a, b) = \begin{cases} (c, d), & \varepsilon = 1, \\ (d, c), & \varepsilon = -1, \end{cases} \quad q = m - hl - t, \quad H = \lceil h/2 \rceil.$$

The two possible values of c^h/a are $K^{(h-1)/2}(c/d)^{(h-1)/2}$ and $K^{(h-1)/2}(c/d)^{(h+1)/2}$. Consequently

$$\lambda^{-(h+1)/2} < 3^{h-1} \frac{c^h}{a} \leq \left(\frac{3}{\sqrt{5}} \right)^{h-1} \lambda^{(h+1)/2}. \quad (26)$$

Since $m \geq 1$, the reciprocal term in the single entry gives

$$1 \leq \frac{u_{\varepsilon m}}{a\lambda^m} \leq 1 + \lambda^{-1} < \frac{3}{2}. \quad (27)$$

Equations (26) and (27), together with $V_l > X_l$, imply

$$\lambda^q > \frac{2}{3} \lambda^{-(h+1)/2} > \lambda^{-(h+2)/2}.$$

The last inequality follows from $\lambda > 5/2 > 9/4$. Thus the integral exponent satisfies $q \geq -H$.

For $l \geq 1$, use (24) for the upper estimate. The inequalities $1 + \lambda^{-1} < 7/5$ and $21/(5\sqrt{5}) < 2 < \lambda$ give $\lambda^q < \lambda^{(3h+1)/2}$. For $l = 0$, the binomial estimate has an extra factor λ^h . Integer rounding, for both parities of h , therefore gives the complete window

$$-\left\lceil \frac{h}{2} \right\rceil \leq q \leq \begin{cases} \lfloor 3h/2 \rfloor, & l \geq 1, \\ \lfloor 5h/2 \rfloor, & l = 0. \end{cases} \quad (28)$$

3.4. The algebraic norm bounds the product index. Rearrange the equality as

$$\lambda^{hl+t} \Delta = \mathcal{R}_l - b\lambda^{-m}, \quad \Delta = a\lambda^q - 3^{h-1}c^h. \quad (29)$$

The two terms defining Δ have norms, in $\mathbb{Q}(\sqrt{D})$, K and $K(9K)^{h-1}$, respectively. They differ because $9K > 1$. Thus $\Delta \neq 0$; this also follows, with its sign, from Lemma 3.2. The norm argument additionally gives a quantitative separation.

The integer D lies strictly between $(3p-1)^2$ and $(3p)^2$, so it is nonsquare. The number λ is an algebraic-integer unit, and $\sqrt{D}c$, $\sqrt{D}d$ are algebraic integers by the exact ray formulas. Hence $(\sqrt{D})^h \Delta$ is an algebraic integer, whether or not D is squarefree. Its nonzero integral norm gives

$$|\Delta \bar{\Delta}| \geq D^{-h}. \quad (30)$$

Conjugation exchanges c, d and replaces λ by λ^{-1} . Using (28), $c, d < \sqrt{p}$, and $H \leq h-1$, we obtain

$$\begin{aligned} |\bar{\Delta}| &< 3^H p^{H+1/2} + 3^{h-1} p^{h/2} \\ &\leq 2 \cdot 3^{h-1} p^{H+1/2}. \end{aligned}$$

Since $D < 9p^2$, equation (30) implies

$$|\Delta| > \frac{1}{2 \cdot 3^{3h-1} p^{2h+H+1/2}}. \quad (31)$$

Meanwhile (25), $b < \sqrt{p}$ and $m \geq 1$ show that the absolute value of the right side of (29) is less than $2^h 3^{h-1} p^{h/2} \lambda^{(h-2)l+t}$. Combining the two estimates and cancelling positive powers of λ gives

$$\lambda^{2l} < 2^{h+1} 3^{4h-2} p^{5h/2+H+1/2} \leq 2^{h+1} 3^{4h-2} p^{3h+1}. \quad (32)$$

The last step uses $H \leq (h+1)/2$. Now

$$2^{h+1} 3^{4h-2} = \frac{2}{9} 162^h, \quad (5/2)^6 = \frac{15625}{64} > 162.$$

If $l \geq 3h$, the lower bound $\lambda > (5/2)p$ contradicts (32), since $6h \geq 3h+1$. Therefore

$$l \leq 3h-1. \quad (33)$$

For $l \geq 1$, equations (18) and (28) give

$$k \leq l + m = (h+1)l + t + q \leq (h+1)(3h-1) + h-1 + \lfloor 3h/2 \rfloor = b(h).$$

For $l = 0$ they give the smaller bound $k \leq h-1 + \lfloor 5h/2 \rfloor$. This proves Theorem 3.1.

The conclusion bounds the gap at fixed h , rather than the occurrence count itself. More importantly for the next sections, it supplies the complete index domain (28)–(33) without imposing a numerical cutoff on the Markov parameter p . The remaining task is to bound that parameter independently of the ray.

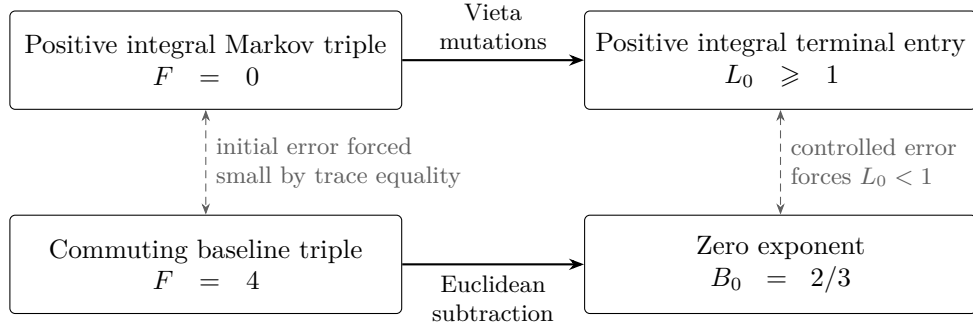


FIGURE 1. The two trace evolutions in the proof. Here $F(X, Y, Z) = X^2 + Y^2 + Z^2 - XYZ$ uses full traces, whereas L_i and B_i are traces divided by three. Both evolutions use Vieta mutation on their own invariant level. Error control compares them without identifying the levels; at the final step it contradicts positive integrality. The corresponding normalized forcing is $4/9$.

4. POSITIVE PRODUCT INDICES AND A UNIFORM VIETA ESTIMATE

At a positive product index, every lower-state visit pays a factor of at least λ^{-1} . This makes the leading discrepancy small. Equality then places an actual integral Markov triple close to a triple of commuting matrix traces. The latter follows the Euclidean algorithm exactly. We follow it with actual Vieta mutations, using the positive quotient formula to keep the error uniformly bounded.

Theorem 4.1 (The positive-index parameter bound). *In the minimum-ray model (15)–(16), let $h \geq 2$, $1 \leq t < h$, and let the gap bits be arbitrary with sum t . If $l \geq 1$ and $V_l = u_{\varepsilon m}$ for an integer $m \geq 1$, then*

$$p < 4^{h+1}. \quad (34)$$

Both signs of the single index are included. The gap pattern need not be mechanical.

4.1. A small positive discrepancy determines rational exponents. Use the quantities from Section 3, and put

$$Q = 3^{h-1}c^h, \quad q = m - hl - t, \quad \delta = \frac{a\lambda^q}{Q} - 1, \quad (a, b) = \begin{cases} (c, d), & \varepsilon = 1, \\ (d, c), & \varepsilon = -1. \end{cases}$$

Lemma 3.2 gives $\delta > 0$. The exact equality also gives $\delta < \mathcal{R}_l/X_l$. A mixed path with j lower positions has at least two transitions. Its relative weight (23) is at most $\sigma\lambda^{-j}$. The all-lower path is bounded separately by λ^{-h-2} , because $l, t \geq 1$. Hence

$$0 < \delta < \frac{4}{9p^2} \{(1 + \lambda^{-1})^h - 1\} + \lambda^{-h-2}. \quad (35)$$

If $p \geq h$, then $h/\lambda < 2/5$. The elementary geometric majorant $\binom{h}{j} \leq h^j$ gives

$$(1 + \lambda^{-1})^h - 1 < \frac{h/\lambda}{1 - h/\lambda} < \frac{5h}{3\lambda}.$$

Since $h \geq 2$, we may bound the last term in (35) by $\lambda^{-4} < 16/(625p^4)$. Therefore

$$0 < \delta < \frac{1}{p^3} \left(\frac{8h}{27} + \frac{16}{625} \right) < \frac{h}{p^3}. \quad (36)$$

For $p \geq 3h$, define

$$e_0 = \frac{4}{D}, \quad \kappa = 9K = 1 + e_0. \quad (37)$$

Then $0 < \delta < 1/(3p^2) < e_0 < 1/8$.

The defining equation $a\lambda^q = (1 + \delta)3^{h-1}c^h$ now fixes the coefficients up to a small multiplicative error. If $a = c$, set $n = h - 1$; if $a = d$, set $n = h + 1$. In either case

$$\begin{aligned} c &= \frac{\lambda^{q/n}}{3}\alpha, & d &= \frac{\lambda^{-q/n}}{3}\beta, & \beta &= \kappa/\alpha, \\ \alpha &= \begin{cases} (1 + \delta)^{-1/n}, & a = c, \\ [\kappa/(1 + \delta)]^{1/n}, & a = d. \end{cases} \end{aligned} \quad (38)$$

These are exact formulas, not asymptotic expansions. In the first case $\alpha = (1 + \delta)^{-1/n}$ satisfies $1 - e_0 < \alpha < 1$, while $\beta = \kappa(1 + \delta)^{1/n}$ satisfies $1 < \beta < (1 + e_0)^2 < 1 + 3e_0$; the upper bound for β comes from $(1 + \delta)^{1/n} \leq 1 + \delta < 1 + e_0$, not from the lower bound on α , which would give only $(1 + e_0)/(1 - e_0)$. In the second case $1 < \alpha, \beta < 1 + e_0$, because $\delta < e_0$. Thus, in both cases,

$$|\alpha - 1|, |\beta - 1| < 3e_0, \quad \max(\alpha/\beta, \beta/\alpha) < 2. \quad (39)$$

If $q = 0$, these estimates would give $u_0 = (\alpha + \beta)/3 < 2(1 + 3e_0)/3 < 1$, contradicting integrality.

Put $s = \lambda^{1/n}$ and $j = |q| > 0$. The minimum ratio in (15) gives

$$s^{2j-n} \leq \begin{cases} \beta/\alpha, & q > 0, \\ \alpha/\beta, & q < 0. \end{cases} \quad (40)$$

For $s \geq 4$, the integer $2j - n$ cannot be positive. Consequently $1 \leq j \leq n/2$. In particular the case $n = 1$, which can occur when $h = 2$, is already impossible. Every remaining case has $n \geq 2$.

Define the commuting baselines

$$B_i(s) = \frac{s^i + s^{-i}}{3}, \quad i \geq 0. \quad (41)$$

We abbreviate these scalar quantities to B_i when s is fixed; they are distinct from the generator matrix B . The actual neighboring Markov triple is

$$x = u_0, \quad z = u_{-\text{sign}(q)}, \quad p = B_n.$$

Its baseline indices are respectively $j, n - j, n$. For example, when $q > 0$, the two terms of z have exponents $-(n - j)$ and $n - j$; when $q < 0$, the same statement holds with α, β interchanged. Thus (39) gives, for both entries,

$$|L_i - B_i| < e_0(s^i + s^{-i}), \quad i = j, n - j, \quad L_n = p = B_n. \quad (42)$$

The factor three in (39) cancels the division by three in the Binet formulas.

4.2. A positive quotient suppresses the mutation error.

Lemma 4.2 (A uniform Vieta neighborhood). *Let $n \geq 2$, $1 \leq j < n$, and $s \geq 4$. There is no positive integral Markov triple with entries labeled by the indices $(j, n - j, n)$ such that*

$$|L_i - B_i(s)| \leq 5s^{i-2n} \quad \text{for each of its three entries.} \quad (43)$$

The baseline indices need not initially be in the order of the actual integer entries.

Proof. For positive indices a, b , put $c = a + b$ and $d = |a - b|$. The commuting baselines satisfy

$$3B_a B_b - B_c = B_d, \quad B_a^2 + B_b^2 - B_c B_d = \frac{4}{9}. \quad (44)$$

Both identities follow by expanding $s^i + s^{-i}$. In particular their Markov defect is $4/9$, which must be retained in any error estimate. For the actual triple, Vieta mutation instead gives

$$L_d = 3L_a L_b - L_c = \frac{L_a^2 + L_b^2}{L_c}. \quad (45)$$

The first formula proves integrality; the second proves positivity. The resulting triple is again Markov. No ordering of the actual entries is needed for these conclusions.

At maximum baseline index $c = a + b$, we maintain

$$|L_i - B_i| \leq 5s^{i-2c} \quad \text{for all three entries.} \quad (46)$$

It holds initially at $c = n$. Exchange a, b if necessary to have $a \geq b$, so $d = a - b$. Write $E_i = L_i - B_i$. Combining (44) and (45) gives the exact error equation

$$L_c(L_d - B_d) = 2B_a E_a + E_a^2 + 2B_b E_b + E_b^2 - B_d E_c + \frac{4}{9}. \quad (47)$$

The quadratic errors are included. Since $c \geq 2$,

$$L_c \geq \left(\frac{1}{3} - 5s^{-2c}\right) s^c \geq \left(\frac{1}{3} - \frac{5}{256}\right) s^c > 0. \quad (48)$$

For every $i \geq 0$, including the possible terminal index zero, $B_i \leq (2/3)s^i$. Equation (46) therefore bounds the numerator in (47) by

$$\begin{aligned} 10s^{-2b} + \frac{20}{3}s^{-2a} + 25s^{-2c}(s^{-2a} + s^{-2b}) + \frac{4}{9} \\ \leq \frac{50}{3}s^{-2} + 50s^{-6} + \frac{4}{9} \\ \leq \frac{50}{48} + \frac{50}{4096} + \frac{4}{9} < 5 \left(\frac{1}{3} - \frac{5}{256}\right). \end{aligned}$$

The last strict margin is $1303/18432 > 0$. Dividing by (48) yields

$$|L_d - B_d| < 5s^{-c} = 5s^{d-2a}. \quad (49)$$

If $a > b$, the new maximum baseline index is $a < c$. The new entry satisfies exactly (46) at that index, and the unchanged entries satisfy its weaker new bounds. Thus the constant five does not grow with the number of mutations.

Repeated subtraction of positive indices terminates at $(g, g, 2g)$, where $g = \gcd(j, n) \geq 1$. Make one further actual mutation. The preceding estimates also hold for $a = b = g$ and $d = 0$, giving

$$\left|L_0 - \frac{2}{3}\right| < 5s^{-2g} \leq \frac{5}{16}.$$

But L_0 is a positive integer and $L_0 < 2/3 + 5/16 = 47/48 < 1$. This contradiction also covers the case $j = n/2$, where the terminal mutation is the first move. $\square$

4.3. Completion of the parameter bound. Suppose $p \geq 4^{h+1}$. Then $p > 3h$, so the coefficient normalization above applies, and for either choice of $n \leq h + 1$,

$$s = \lambda^{1/n} > 4^{(h+1)/n} \geq 4.$$

If $n = 1$ there is already a contradiction in (40). Otherwise $n \geq 2$, and the exact identity $\sqrt{D} = \lambda - \lambda^{-1}$ gives

$$e_0 = \frac{4s^{-2n}}{(1 - s^{-2n})^2}.$$

For $i \geq 1$, $s \geq 4$ and $n \geq 2$,

$$e_0(s^i + s^{-i}) \leq \frac{17}{16} \frac{4}{(1 - 1/256)^2} s^{i-2n} = \frac{16384}{3825} s^{i-2n} < 5s^{i-2n}. \quad (50)$$

Thus (42) gives the prohibited triple of Lemma 4.2. This proves Theorem 4.1.

The role of integrality is concentrated at the end of the argument. The commuting baselines follow the subtractive Euclidean algorithm to $B_0 = 2/3$. Positive quotient mutation keeps an actual Markov orbit close enough that its terminal entry would also lie below one. This controls all positive product indices with a single error constant, rather than estimating a growing sequence of differences of large traces.

5. MECHANICAL MINIMUM PATTERNS AND SPECTRAL DESCENT

When the product index is zero, the estimate used in the preceding section loses its factor λ^{-1} per lower-state visit. A different input is needed. Cyclic balance controls each contiguous block of lower states, and the equality itself turns this control into an integer gap in the block exponent. We then measure the Vieta error in the spectral coordinate $\text{arcosh}(3L/2)$. In that coordinate a mutation is almost a subtraction, and the integer coefficients of the Euclidean algorithm control the accumulated error.

Theorem 5.1 (The minimum-index parameter bound). *In the minimum-ray model (15)–(16), let $h \geq 2$ and let $\epsilon_1, \dots, \epsilon_h$ be a cyclic mechanical binary pattern with $1 \leq t = \sum_i \epsilon_i < h$. If*

$$V_0 = \frac{1}{3} \text{tr} \prod_{i=1}^h A_0 B^{\epsilon_i} = u_{\epsilon m}, \quad m \geq 1, \quad (51)$$

then

$$p < (100h^2)^{h+1}. \quad (52)$$

The analytic statement does not require $\gcd(h, t) = 1$.

Here cyclic mechanical means that, for some intercept ω , the bits are successive differences of $\lfloor it/h + \omega \rfloor$, extended periodically with period h . For every proper cyclic interval of j positions, its number e of ones therefore satisfies

$$|e - tj/h| < 1. \quad (53)$$

Indeed the difference from tj/h is the difference of the fractional parts at the two endpoints. This strict interval estimate is the only mechanical-word property used in the proof.

5.1. Choose a dominant state before estimating the error. Set $\kappa = 9K = 1 + e_0$, where $e_0 = 4/D$, as in (37). Choose the larger of the two constant-state contributions to V_0 . If it is the upper contribution, put $\gamma = 1$ and $(c_0, d_0) = (c, d)$. Otherwise interchange the two matrix coordinates, put $\gamma = -1$ and $(c_0, d_0) = (d, c)$, and use signed gaps $\gamma\epsilon_i$ with the displayed diagonal matrix B . This operation leaves the word trace unchanged. The represented ray is now u_i or u_{-i} respectively, and the signed single index is $\gamma\epsilon m$. In particular its magnitude and minimum normalization are unchanged. These transformations are changes of the trace model; no surface isometry is asserted.

Put

$$\begin{aligned} \tau &= \gamma t, & X &= 3^{h-1} c_0^h \lambda^\tau, & C &= K \kappa^{h-1}, \\ \rho &= d_0/c_0, & Z &= C/X^2 \leq 1, & \sigma &= 4/(9p^2). \end{aligned} \quad (54)$$

The other constant contribution is C/X . If $\mathcal{M}$ denotes the sum of all mixed path contributions, then

$$V_0 = X(1 + Z) + \mathcal{M}. \quad (55)$$

For a mixed path, the proper set of lower positions consists of r nonempty contiguous cyclic blocks. Let j be its total size and e its total number of original one bits. The path expansion of Section 3 gives the exact relative weight

$$w = \sigma^r \rho^j \lambda^{-2\gamma e}. \quad (56)$$

For a block of size j_i and one-count e_i , the number $1 + \gamma(e_i - tj_i/h)$ is positive by (53). It is a multiple of $1/h$, and hence at least $1/h$. Since $Z = \rho^h \lambda^{-2\gamma t}$, multiplying these block estimates in (56) gives

$$w \leq a_0^r z_0^j, \quad a_0 = \sigma \lambda^2 \lambda^{-2/h} < 4\lambda^{-2/h}, \quad z_0 = Z^{1/h} \leq 1. \quad (57)$$

The exponent gain is paid once per lower block, rather than once per position. This distinction will also control the number of relevant paths.

5.2. Sum paths by their transitions. For positive scalars a, z , consider

$$T(a, z) = \begin{pmatrix} 1 & \sqrt{az} \\ \sqrt{az} & z \end{pmatrix}, \quad \Phi_h(a, z) = \text{tr } T(a, z)^h - 1 - z^h.$$

A mixed cyclic state path with j lower positions and $2r$ transitions contributes exactly $a^r z^j$ to this trace. All coefficients are positive. Choosing its $2r$ transition edges and an initial state gives exactly $2\binom{h}{2r}$ such paths when the lower-state size is not prescribed. It follows that

$$\begin{aligned} \Psi_h(a) &:= \Phi_h(a, 1) = 2 \sum_{r=1}^{\lfloor h/2 \rfloor} \binom{h}{2r} a^r \\ &= (1 + \sqrt{a})^h + (1 - \sqrt{a})^h - 2. \end{aligned} \quad (58)$$

For $h^2 a < 1$, the inequality $2\binom{h}{2r} \leq h^{2r}$ gives the useful rational bound

$$\Psi_h(a) \leq \frac{h^2 a}{1 - h^2 a}. \quad (59)$$

Thus (57) gives, before any small-discrepancy assumption,

$$\mathcal{M}/X \leq \Phi_h(a_0, z_0) \leq \Psi_h(a_0). \quad (60)$$

This is a sum of positive indexed paths, so cyclic repetitions or a nonprimitive period do not change its validity.

Suppose henceforth, for a contradiction, that

$$p \geq S^{h+1}, \quad S = 100h^2. \quad (61)$$

Then $\lambda > S^{h+1}$ and $h^2 a_0 < 4h^2/S^2 < 1/16$. Equations (59) and (60) imply

$$\mathcal{M}/X < \frac{64h^2}{15S^2} < \frac{1}{16}. \quad (62)$$

Let a be the leading coefficient of the single entry in the transformed ray, so a is c_0 or d_0 , and put $Y = a\lambda^m$. Lemma 3.2 gives $Y > X$. The minimum ratio also gives $Y > \sqrt{\lambda}/3$. By (55), $Z \leq 1$ and (62), we have $V_0 < 3X$; hence $X > \sqrt{\lambda}/9$. Since $C < 2^h$ and $S \geq 400 > 32$,

$$Z < \frac{81 \cdot 2^h}{\lambda} < \frac{81}{32 \cdot 16^h} \leq \frac{81}{8192} < \frac{1}{16}. \quad (63)$$

The exact equality $Y + K/Y = V_0$ now shows that

$$0 < \delta := Y/X - 1 < Z + \mathcal{M}/X < \frac{1}{8}. \quad (64)$$

The opposite constant path was not discarded: its smallness follows only after using equality and the integral minimum normalization.

5.3. Balance becomes a gap in an integer exponent. Define

$$q = m - \tau, \quad n = \begin{cases} h - 1, & a = c_0, \\ h + 1, & a = d_0. \end{cases} \quad (65)$$

This internal offset may differ from $m - t$, because the dominant-state choice can have $\gamma = -1$. The original enumeration convention of (28) is unchanged. As in (38), exact rearrangement gives

$$\begin{aligned} c_0 &= \frac{\lambda^{q/n}}{3}\alpha, & d_0 &= \frac{\lambda^{-q/n}}{3}\beta, & \beta &= \kappa/\alpha, \\ \alpha &= \begin{cases} (1 + \delta)^{-1/n}, & a = c_0, \\ [\kappa/(1 + \delta)]^{1/n}, & a = d_0. \end{cases} \end{aligned} \quad (66)$$

Here $e_0 = 4/D < 1/8$, but there is no assumption that $\delta < e_0$. Positive-root monotonicity gives

$$\begin{aligned} 8/9 < \alpha < 9/8, & \quad 1 < \beta < 81/64, \\ |\alpha - 1|, |\beta - 1| < 2(e_0 + \delta), & \quad \max(\alpha/\beta, \beta/\alpha) < 3/2. \end{aligned} \quad (67)$$

For completeness, α lies between $1/(1 + \delta)$ and κ , and $1 < \beta \leq \kappa(1 + \delta)$. These bounds imply the displayed error estimates; the ratio bound follows from $(81/64)/(8/9) = 729/512 < 3/2$. If $q = 0$, then $u_0 = (\alpha + \beta)/3 < 51/64 < 1$, impossible.

Set

$$s = \lambda^{1/n} > S, \quad j_0 = |q| > 0, \quad g = \gcd(n, j_0). \quad (68)$$

The minimum ratio and (67) give $s^{2j_0-n} < 3/2$ whenever $2j_0 - n$ is positive. Since $s > 4$, integrality implies $1 \leq j_0 \leq n/2$. As in the preceding section, $n = 1$ is thereby excluded immediately. The remaining cases have $n \geq 2$, and $2h/n \leq 3$: this is immediate for $h \geq 3$; at $h = 2$ the only remaining choice is $n = 3$.

For each proper lower block define the integers

$$F_i = n + qj_i + \gamma ne_i, \quad E = hq + n\tau = hm + (n - h)\tau. \quad (69)$$

Since $m \geq 1$, $n - h = \pm 1$ and $\tau = \pm t$, $E \geq h - t > 0$. Moreover

$$F_i = n\{1 + \gamma(e_i - tj_i/h)\} + Ej_i/h > 0. \quad (70)$$

Both E and every F_i are multiples of g , so $E \geq g$ and $\sum_i F_i \geq rg$. This is where the actual equality improves the coarse denominator h in (57) to the integer divisor g .

Put $R = \beta/\alpha > 1$ and $a_g = \sigma\lambda^2 s^{-2g}$. Rewriting the exact weight gives

$$w = (\sigma\lambda^2)^r R^j s^{-2\sum_i F_i} \leq a_g^r R^j, \quad Z = R^h s^{-2E}. \quad (71)$$

The coefficient normalization also gives

$$R = \kappa^\zeta (1 + \delta)^{2/n}, \quad \zeta = \begin{cases} 1, & a = c_0, \\ 1 - 2/n, & a = d_0. \end{cases}$$

In the case $a = d_0$ one has $n = h + 1 \geq 3$ by (65), so $1/3 \leq \zeta \leq 1$ in both cases. Since $he_0 \leq 4h/(5p^2) < 1/2$, the binomial/geometric estimate yields $\kappa^h < 1/(1 - he_0) < 2$. Together with $2h/n \leq 3$ this gives

$$R^h < 2(9/8)^3 < 3. \quad (72)$$

Equations (59), (71) and (72), with $h^2 a_g < 4h^2/S^2 < 1/16$, imply

$$0 < \delta < 3s^{-2g} + 3\Psi_h(a_g) < \left(3 + \frac{64}{5}h^2\right)s^{-2g}. \quad (73)$$

Also $e_0 = 4s^{-2n}/(1 - s^{-2n})^2 < 5s^{-2n}$. Thus

$$e_0 + \delta < \left(8 + \frac{64}{5}h^2\right)s^{-2g} \leq 15h^2s^{-2g}, \quad (74)$$

where the last inequality holds for every $h \geq 2$.

Take the actual ray neighbor of u_0 at index $-\text{sign}(q)$ in the transformed ray. These two entries and p form a positive integral Markov triple with baseline indices $(j_0, n - j_0, n)$. Using the scalar baselines (41), the exact coefficients give

$$L_n = p = B_n, \quad |L_i - B_i| < 20h^2s^{i-2g}, \quad i = j_0, n - j_0. \quad (75)$$

Unlike (42), this error is measured at the fixed divisor scale g , rather than at the initial maximum index n . Direct propagation of relative errors would therefore lose a factor at each mutation. Spectral coordinates avoid that loss.

5.4. The exact spectral error of a Vieta mutation. For a real label $L > 2/3$, define

$$\theta(L) = \text{arcosh}(3L/2). \quad (76)$$

Then $\theta(B_i) = i \log s$ for $i > 0$. Let L_a, L_b, L_c be a positive integral Markov triple, with L_c strictly largest, and put $A = \theta(L_a)$, $B' = \theta(L_b)$. Solving the quadratic in the third full trace gives

$$\frac{3}{2}L_c = \cosh A \cosh B' + \sqrt{\sinh^2 A \sinh^2 B' - 1}. \quad (77)$$

The Vieta partner $L_d = 3L_aL_b - L_c$ is therefore determined by

$$\begin{aligned} \frac{3}{2}L_d &= \cosh(A - B') + \epsilon(A, B'), \\ \epsilon(A, B') &= \frac{1}{\sinh A \sinh B' + \sqrt{\sinh^2 A \sinh^2 B' - 1}}. \end{aligned} \quad (78)$$

The square root is real and positive: $L_a, L_b \geq 1$ give $\sinh^2 A, \sinh^2 B' \geq 5/4$. The sign in (77) is forced by the largest-entry hypothesis. Indeed, if $x \geq y \geq 3$ are the other full traces, then the trace quadratic evaluated at $z = x$ is

$$x^2(2 - y) + y^2 \leq y^2(3 - y) \leq 0.$$

A third trace strictly greater than x must be its larger root. The expression for ϵ follows by rationalizing the difference $\sinh A \sinh B' - \sqrt{\sinh^2 A \sinh^2 B' - 1}$.

At a nonterminal baseline stage let $a > b \geq g$, $c = a + b$, and $d = a - b$. Suppose all three angle errors from their baselines are less than $1/8$. Since each distinct-index gap is at least $g \log s$, the actual entries have the same strict ordering as their baseline indices. In particular (77) uses the correct root, and $A > B'$. For $i \geq g$, $|v| \leq 1/4$ and $s \geq 4$,

$$\sinh(i \log s - v) \geq \frac{3}{8}s^i(1 - 2s^{-2i}) \geq \frac{21}{64}s^i > \frac{1}{4}s^i. \quad (79)$$

Here $\exp(-1/4) > 3/4$ and $\exp(1/2) < 2$, as follows from their power series. Apply this estimate to A , B' and $A - B'$. The inverse-cosh mean-value estimate in (78) then gives an exact additive error relation

$$\theta(L_d) = A - B' + \eta, \quad 0 < \eta < 64s^{-2a} \leq 64s^{-4g}. \quad (80)$$

Indeed $\eta \leq \epsilon(A, B')/\sinh(A - B')$, and the three $\sinh$ factors are bounded below by $s^a/4$, $s^b/4$, $s^{a-b}/4$. Equation (80) is the reason to use spectral rather than relative label errors.

5.5. Euclidean rows give a polynomial error bound.

Lemma 5.2 (Coefficients along subtractive Euclid). *Start with positive integer indices a, b of $\gcd g$, and put $N = (a + b)/g$. Under repeated subtraction of the smaller index from the larger, every signed coefficient row expressing a current index in terms of the starting pair has ℓ^1 norm at most N . The same bound applies to propagation from any intermediate stage. If the two initial coordinate errors are bounded by e_* and every additive update error is bounded by f_* , then all later coordinate errors are bounded by*

$$Ne_* + N(N - 1)f_* < Ne_* + N^2f_*. \quad (81)$$

Proof. Track the two coefficient rows r, s , initially $(1, 0), (0, 1)$. They remain unimodular. In each coordinate i , the weak opposite-sign condition $r_i s_i \leq 0$ is preserved: replacing r by $r - s$ changes the product to $r_i s_i - s_i^2 \leq 0$. Consequently $|r - s| = |r| + |s|$ componentwise. Each new absolute row dominates both rows that produced it.

Include the final subtraction of equal indices, producing zero and g . If $A_* = a/g$ and $B_* = b/g$, the zero row is a primitive integer relation on the coprime pair (A_*, B_*) , hence is $\pm(B_*, -A_*)$. It dominates every preceding absolute row and has ℓ^1 norm N . This proves the row bound. At any intermediate stage the $\gcd$ is still g , while the normalized sum is at most N , so the same proof applies to every remaining tail. There are at most $N - 1$ subtractions, since each decreases the normalized sum by at least one. Propagating each additive error along its remaining tail and summing proves (81). Coordinate exchanges have norm one and require no additional estimate. $\square$

We now apply the lemma to the angle errors, with initial nonlargest indices $j_0, n - j_0$. The actual index- n entry has zero initial error. From (75), every point on the segment between L_i and B_i is at least $s^i/4$, because $20h^2s^{-2g} < 1/12$. On this segment

$$\theta'(L) = \frac{1}{\sqrt{L^2 - 4/9}} < 8s^{-i}.$$

The initial angle errors are therefore bounded by $e_* = 160h^2s^{-2g}$. We have $N = n/g \leq h + 1 \leq 2h$. If the errors stay below $1/8$, equation (80) permits $f_* = 64s^{-4g}$. The bound from Lemma 5.2 is then

$$Ne_* + N^2f_* < \frac{320h^3}{S^2} + \frac{256h^2}{S^4} \leq \frac{2}{125} + \frac{256}{10^8} < \frac{1}{16}. \quad (82)$$

The constant $2/125$ includes the boundary $h = 2$.

This proves the assumed error bound by a first-violation argument. Initially all errors are below $1/8$. Before a first violating move, the previously proved ordering selects the larger root at every stage, and every forcing is bounded by (80). The row estimate then bounds the new error by (82), a contradiction. Only the two nonlargest baseline coordinates are updated as a subtractive pair. The third entry is initially exact, and thereafter is an unchanged entry from the previous pair, so it has the same error bound. There is no extra independent error from a discarded large coordinate.

5.6. The terminal contradiction. The subtractive Euclidean algorithm reaches $(g, g, 2g)$. The two index- g angles then satisfy $|A - B'| < 1/8$, while each is at least $g \log s - 1/16$. By (79), $\epsilon(A, B') < 16s^{-2g} < 1/16$. The actual index- $2g$ entry is still strictly largest, so its positive integral Vieta partner satisfies (78). Using $\cosh(1/4) < \sum_{j \geq 0} (1/4)^{2j} = 16/15$, we obtain

$$0 < L_d < \frac{2}{3} \left(\frac{16}{15} + \frac{1}{16} \right) = \frac{271}{360} < 1.$$

This contradicts positive integrality. It includes the equal-index starting case $j_0 = n/2$, with no nonterminal steps. Together with the earlier direct exclusion of $n = 1$, it proves Theorem 5.1.

Two different estimates were needed. The first used cyclic balance before any approximation assumption to select a small discrepancy. The second used the exact equality to make all block exponents positive multiples of g . Spectral subtraction then transported the resulting error with polynomial coefficients. The theorem supplies a finite parameter range at every prescribed occurrence count; it does not assert that a constant-state path dominates uniformly when h grows on a fixed ray. The next section turns the two parameter bounds into a complete finite comparison and isometry decision.

6. A COMPLETE FINITE DECISION AT EACH OCCURRENCE COUNT

The parameter bounds become useful only when accompanied by an exhaustive and economical way to generate the relevant markings. Scanning all integers below the height bound would obscure the main gain: the height is exponential in h , but the number of Markov occurrences below it is polynomial in h . We first establish this distinction, then explain how to recognize the geometric meaning of every equality.

6.1. Generating all minimum seeds.

Lemma 6.1 (Minimum seeds and ordered occurrences). *Every minimum-centered positive integral Markov ray with fixed parameter p has one of the two initial pairs*

$$(u_0, u_1) = (a, b) \quad \text{or} \quad (a, 3pa - b), \quad (83)$$

where (a, b, p) is an ordered positive Markov triple. Conversely, each such triple supplies both rays in (83), with u_0 a minimum. All ordered occurrences with maximum at most H are generated by the two initial occurrences $(1, 1, 1), (1, 1, 2)$ and the tree rooted at $(1, 2, 5)$ with children

$$(a, b, c) \mapsto (a, c, 3ac - b), \quad (b, c, 3bc - a), \quad (84)$$

pruning a child when its maximum exceeds H .

Proof. Let $x = u_0$, $z = \min(u_{-1}, u_1)$, and let y be the other neighbor. They are roots of $f(T) = T^2 - 3pxT + p^2 + x^2$ and satisfy $x \leq z$. The minimum bound implies $x \leq p$, while

$$f(p) = x^2 - (3x - 2)p^2 \leq 0.$$

Thus $z \leq p \leq y$, and (x, z, p) is an ordered Markov triple. Conversely, for an ordered triple (a, b, p) the other neighbor is

$$3pa - b = \frac{p^2 + a^2}{b} \geq p \geq a.$$

Both immediate neighbors are at least a ; the recurrence, or Lemma 2.1, then makes a a global minimum. The two orientations are distinct: equality of the neighbors would give $b^2 = p^2 + a^2 > p^2$, contradicting $b \leq p$.

For completeness of the tree, consider an ordered triple $a \leq b \leq c$. The conjugate root $c' = 3ab - c = (a^2 + b^2)/c$ is positive. The polynomial $g(T) = T^2 - 3abT + a^2 + b^2$ satisfies

$$g(b) = a^2 - (3a - 2)b^2 \leq 0,$$

with equality only if $a = b = 1$. Outside $(1, 1, 1)$ and $(1, 1, 2)$, we have $0 < c' < b < c$. Replacing the largest entry and reordering therefore gives a uniquely determined parent of smaller height. Iteration reaches the initial occurrences. Equal coordinates can occur only there: descent preserves the multiset of pairwise greatest common divisors, which are all one at the initial triples. Thus $a = b$ forces $a = b = 1$, while $g(b) < 0$ excludes

$b = c$ outside the initial cases. Reversing this unique descent gives exactly (84), with strictly increasing maximum. No uniqueness assertion concerning triples with the same maximum is used. $\square$

The construction has a literal matrix realization. Set $C = EFE = \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix}$. For the three initial occurrences use, respectively,

$$(U, V) = (E, F), \quad (E, FE), \quad (E, C).$$

The normalized traces of (U, V, UV) are the displayed triples. The two child operations are

$$(U, V) \mapsto (U, UV), \quad (V, UV). \quad (85)$$

At a retained occurrence (a, b, p) , choose $(A_0, B) = (U, (UV)^{-1})$ or (U, UV) . These realize exactly (83). Starting from the root columns $[E] = (1, 0)$, $[F] = (0, 1)$, update the columns by addition in (85), and negate the second one when inverting UV . Thus the representation and its marking are generated together.

This model loses no marked realization. Diagonalizing its second generator determines the diagonal entries of the first from the complete trace triple; its off-diagonal product is positive and fixed by the determinant. The real conjugation in Lemma 2.1 therefore identifies any two such full characters. It conjugates the whole group, so also transports its isometries. Agreement of just one matrix trace would not provide this conclusion.

6.2. Counting occurrences rather than integer labels.

Lemma 6.2 (A quadratic count in word length). *If every retained positive Cohn word has length at most J , the seed construction has at most $J(J-1) + 4$ oriented seeds. A word of length j in E, C has trace strictly larger than $(5/2)^j$.*

Proof. The positive tree in (85), starting at (E, C) , assigns each new maximum a positive primitive count vector (r, s) . These vectors are distinct before traces are evaluated. Indeed adjacent vectors have determinant of absolute value one, and inserting their sum splits the interval between their slopes into two disjoint open intervals. A later insertion stays strictly inside its own interval. Induction gives the usual Farey mediant tree, without repeated interior slopes. At length $j = r + s$ there are at most $j - 1$ vectors. Summing and including the two initial occurrences gives at most $2 + J(J-1)/2$ occurrences; each has two orientations.

Entrywise, $C \geq E > 0$. Replacing every C by E can only decrease a positive product and its trace. The expanding eigenvalue of E is $(3 + \sqrt{5})/2 > 5/2$, so $\text{tr}(E^j) > (5/2)^j$. $\square$

Theorem 6.3 (Explicit complete decision domain). *Fix $h \geq 2$. All equalities $\text{tr } A = \text{tr } W_{h,k,t}(A, B)$, over every marked modular-torus basis, $k \geq 1$, and $1 \leq t < h$ coprime to h , can be classified by a finite algorithm. Define*

$$\begin{aligned} H_+ &= 4^{h+1} - 1, & J_+ &= 2h + 3, & S_+ &= J_+(J_+ - 1) + 4, \\ H_0 &= (100h^2)^{h+1} - 1, & L &= \lceil \log_2(100h^2) \rceil, \\ J_0 &= (h+1)L + 1, & S_0 &= J_0(J_0 - 1) + 4. \end{aligned} \quad (86)$$

At most

$$\begin{aligned} N_+(h) &= 2S_+(3h-1)(h-1)(2h+1), \\ N_0(h) &= 2S_0(h-1)(3h+1) \end{aligned} \quad (87)$$

exact trace comparisons suffice. Their sum is $O(h^5)$, using integers with $O(h^3)$ bits. Every equal pair can also be tested for actual isometry using twelve fixed integral homology actions. The complexity is polynomial in the numerical value of h .

Construction of the domain. Apply the centering and reversal of Section 3. The positive-product branch has $1 \leq l \leq 3h - 1$ and $p \leq H_+$ by Theorem 4.1. The minimum-product branch has $l = 0$ and $p \leq H_0$ by Theorem 5.1. Generate every seed up to the relevant height by Lemma 6.1. For each seed and each primitive mechanical pattern, compare its word trace with both signed ray entries $u_{\varepsilon m}$, where

$$\begin{array}{c|c|c} & m = hl + t + q \geq 1 & \text{range of } q \\ \hline l \geq 1 & & -\lceil h/2 \rceil \leq q \leq \lfloor 3h/2 \rfloor \\ l = 0 & & -\lceil h/2 \rceil \leq q \leq \lfloor 5h/2 \rfloor. \end{array} \quad (88)$$

These are the original phase windows of Section 3. In particular, the q used here is not changed by the internal spectral-state interchange in the minimum-product proof.

The domain is a convenient superset. To recover exactly the original positive-gap family, put $n = \varepsilon m$ and $d = l - n$. If $d \geq 1$, retain $(k, t) = (d, t)$. If $d \leq -2$, reverse the ray and retain $(k, t) = (-d - 1, h - t)$. Discard $d = 0, -1$, which give no original $k \geq 1$ representation by this normalization. No equalities are removed because their trace or seed label has occurred before.

For the positive branch, a retained maximum word cannot have length $j \geq J_+ + 1 = 2h + 4$, since

$$(5/2)^{2h+4} > 3 \cdot 4^{h+1} > 3H_+.$$

For the minimum branch, $j \geq J_0 + 1$ would give

$$(5/2)^j > 2^j \geq 4(100h^2)^{h+1} > 3H_0.$$

Lemma 6.2 gives S_+, S_0 . There are at most $h - 1$ patterns, two signs, and respectively $2h + 1$ and $3h + 1$ offsets. This proves (87). One may replace $h - 1$ by Euler's totient $\varphi(h)$ when estimating the size of the domain more sharply.

For completeness, seed generation itself has polynomial cost. Each retained node creates at most two first rejected children, whose labels are $O(H^2)$. It never explores their descendants. Every entry of a positive Cohn matrix is at most its trace: the first factor gives row dominance and the last gives column dominance. Seed matrices and their inverses therefore have entries $O(H)$. In the positive branch, all needed exponents are $O(h^2)$ and $\log H_+ = O(h)$, so literal products and recurrence values have $O(h^3)$ bits. In the minimum branch the corresponding bounds are $O(h)$ and $O(h \log h)$, giving $O(h^2 \log h)$ bits. Exact multiplication adds polynomial overhead per comparison. Finally $N_+ = O(h^5)$ and $N_0 = O(h^4 \log^2 h) = O(h^5)$. The promised isometry test is established next. $\square$

6.3. Recognizing every actual isometry.

Proposition 6.4 (The complete homology action). *In the root marking (1), the homology image of the full isometry group of T is exactly*

$$\mathcal{J} = \{\pm G^i, \pm G^i D_0 : i = 0, 1, 2\}, \quad G = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \quad D_0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (89)$$

Two primitive simple classes with root homology columns v, w are isometric if and only if $w = Mv$ for some $M \in \mathcal{J}$.

Proof. The real matrices

$$K_0 = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}, \quad R_0 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \Sigma_0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

normalize the group generated by E, F . Direct multiplication gives $K_0 E K_0^{-1} = F$, $K_0 F K_0^{-1} = (EF)^{-1}$; $R_0 E R_0^{-1} = F$, $R_0 F R_0^{-1} = E$; and $\Sigma_0 E \Sigma_0^{-1} = E^{-1}$, $\Sigma_0 F \Sigma_0^{-1} = F^{-1}$. The negative determinant of R_0 realizes an orientation-reversing hyperbolic isometry.

The resulting homology actions are $G, D_0, -I$, which generate the twelve distinct matrices in (89).

To see that the list is complete, use the shortest simple curves. They are exactly the three classes with normalized trace one, whose signed primitive columns are

$$\mathcal{S} = \{\pm(1, 0), \pm(0, 1), \pm(1, 1)\}.$$

Indeed the initial Farey triangle has these three labels, and every outward step replaces an entry by a new larger maximum. No later vertex introduces another label one. An isometry permutes $\mathcal{S}$. An integral automorphism preserving $\mathcal{S}$ has at most six choices for the image of $(1, 0)$. Once that image is fixed, it has at most two choices for the image of $(0, 1)$: after one of the displayed actions returns the first image to $(1, 0)$, the condition that the sum of the images also belong to $\mathcal{S}$ leaves only $(0, 1)$ and $(-1, -1)$. Thus there are at most twelve actions, all already realized. Finally primitive homology determines the simple class up to orientation, and $-I$ is included. This proves the orbit criterion. $\square$

In a generated seed with columns α, β , an equality row has columns

$$v = \alpha + n\beta, \quad w = h\alpha + (hl + t)\beta. \quad (90)$$

They are primitive because $\det(\alpha, \beta) = \pm 1$ and $\gcd(h, t) = 1$. Their coordinates have size $O(h^3)$: seed columns grow with their Cohn word length, and the coefficients in (90) are $O(h^2)$. The twelve comparisons therefore use $O(\log h)$ -bit coordinates. This completes the algorithm in Theorem 6.3 and distinguishes an equal trace from an equal trace explained by a surface isometry.

7. RIGIDITY AT SMALL OCCURRENCE COUNT AND INTERSECTION

The preceding bounds are effective for every fixed occurrence count. Their first applications give a geometric conclusion: throughout the small families considered here, equal length comes from a symmetry of the surface. The computation enters only to finish a domain that has already been proved finite and complete, and to exhibit the symmetry of every pair that survives it.

Theorem 7.1 (Small occurrence counts). *Let (A, B) be any positively lifted simple basis of the modular punctured torus. Let $2 \leq h \leq 6$, $k \geq 1$, and $1 \leq t < h$ with $\gcd(h, t) = 1$. If*

$$\mathrm{tr} A = \mathrm{tr} W_{h,k,t}(A, B),$$

then an isometry of the modular torus carries the primitive class of A to that of $W_{h,k,t}(A, B)$.

Proof. Theorem 6.3 supplies every minimum seed, product index, pattern, and signed single index required for the equality. The exact evaluations of this domain are summarized in Table 1. Each equality record retains its full marked character and both primitive homology vectors in the standard modular basis. In every record one of the twelve matrices of Proposition 6.4 maps the first vector to the second. This is an actual surface isometry, since the matrices are the homology actions of the explicitly realized modular normalizers. Primitive homology determines the unoriented simple class.

This is a finite assertion about integers, and we verified it by computer in exact arithmetic. The verification was carried out twice, independently: once by multiplying the integral matrices of each seed, and once by generating the Markov tree afresh and evaluating the word in the universal character algebra, so that the second calculation shares neither the matrix arithmetic nor the traversal of the first. Section A.1 records the domain, the orientation conventions and the two trace formulas, and the equality records themselves are in the supplementary material. What makes the conclusion hold

for every marked character and every k , rather than for labels below some threshold, is the domain theorem rather than the computation. $\square$

h	Realized original types (k, t)	Oriented seeds	Comparisons
2	(1, 1), (3, 1)	(12, 134)	2 208
3	(2, 1), (2, 2), (4, 1)	(20, 292)	14 408
4	(1, 1), (1, 3), (3, 1), (3, 3), (5, 1)	(26, 522)	35 352
5	(1, 2), (1, 3), (4, 1), (4, 4), (6, 1)	(32, 836)	136 400
6	(1, 1), (5, 1), (5, 5), (7, 1)	(46, 1 238)	127 324

TABLE 1. The realized types and the size of each finite domain. Each seed pair lists the counts for $l \geq 1$ and $l = 0$, respectively; no equal numerical maxima are merged. Every listed type occurs, and every equality has an explicit isometry witness. Types not listed have no equality in any marked modular character.

The theorem concerns an infinite family for each h , and its finite output need not consist of a single orbit or a single trace. Nor does it assert that every word with the same two letter counts is simple. The mechanical pattern and the full marked character are essential to its scope.

For $h = 2, 3, 4, 5, 6$ respectively there are 6, 8, 13, 13, 10 equalities with an original quotient $k \geq 1$, counted with the duplicate descriptions that tied minima produce. The full characters realizing each type in Table 1, together with the isometry carrying one class to the other, are listed in the supplementary material. Thus the table gives a complete classification despite the unbounded range of possible traces. The count $h = 6$ is not consumed by the application in the next subsection, which meets only $h \leq 5$; it is proved here because the classification is natural to state for the whole range the finite decision reaches.

7.1. An integral reflection criterion. A second mechanism covers certain residues at arbitrarily large intersection. It uses the topology of equal-length loci, rather than a bound on Markov parameters.

Lemma 7.2 (Reflection residues). *Let two distinct primitive unoriented simple classes on a once-punctured torus have vectors $(1, 0)$ and (r, N) , with $N > 0$ and $\gcd(r, N) = 1$. If $r^2 \equiv 1 \pmod{N}$, then on every complete finite-area hyperbolic metric, equality of their lengths forces an orientation-reversing isometry exchanging them.*

Proof. The unique rational map interchanging the two displayed vectors is

$$R = \begin{pmatrix} r & (1 - r^2)/N \\ N & -r \end{pmatrix}. \quad (91)$$

It satisfies $R^2 = I$ and $\det R = -1$. The congruence makes it integral, hence an involution in the extended mapping class group. It exchanges the two simple classes because primitive homology determines them.

We use the fixed-boundary theorem of McShane and Parlier: the equal-length locus of two distinct simple classes on a one-holed torus, at a fixed boundary length, is an embedded path with both ends on the Thurston boundary [11, Theorem 1.3]. Their convention includes a cusp as boundary, so the theorem applies at boundary length zero. In particular, the locus $\mathcal{E}$ inside punctured-torus Teichmüller space is a connected embedded one-manifold without boundary. This fixed-boundary assertion is stronger than what would follow from merely intersecting an ambient connected hypersurface with

a slice; it is the precise property used here. We do not need the theorem's description of the endpoints. It should be said that the same authors prove that the surfaces failing the Markov isometry property are dense once the boundary length is allowed to vary, and conclude from this that the uniqueness conjecture is probably not tractable by hyperbolic geometry alone [11, Theorem 1.4]. Nothing here contradicts that: the input used above is the fixed-boundary statement at boundary length zero, and it produces an isometry only for the residues satisfying the congruence hypothesis, the remaining residues being handled by the arithmetic of the preceding sections.

Identify this Teichmüller space with the upper half-plane, using the conformal compactification to a marked elliptic curve. In the equivariant period coordinate, the involution acts by

$$\tau \mapsto \frac{r\bar{\tau} + (1 - r^2)/N}{N\bar{\tau} - r}.$$

Its fixed locus is the semicircle

$$\mathcal{F} = \left\{ x + iy : (x - r/N)^2 + y^2 = N^{-2}, y > 0 \right\}.$$

This is a nonempty closed embedded line in the upper half-plane. A fixed conformal structure realizes R by an anticonformal automorphism; uniqueness of the complete hyperbolic metric in that structure makes it an isometry. Hence $\mathcal{F} \subset \mathcal{E}$.

A nonempty closed embedded one-manifold contained in a connected embedded one-manifold is also open there. Indeed a small parametrized interval in the first manifold maps continuously and injectively into a coordinate interval of the second; strict monotonicity gives a neighborhood of its central point. Applied here, this shows that $\mathcal{F}$ is both open and closed in $\mathcal{E}$, so $\mathcal{F} = \mathcal{E}$. Every equal-length structure therefore realizes the required isometry. $\square$

This criterion does not classify all isometries relating equal-length pairs. For example, the order-three isometry at intersection nineteen in Proposition A.1 belongs to another mechanism. The congruence is used as a sufficient condition, and no failed congruence is used to discard a possible equality.

7.2. The complete cover through intersection twenty-four.

Theorem 7.3 (Small intersections). *Two essential simple closed geodesics of equal length on the modular punctured torus, with geometric intersection at most twenty-four, are related by an isometry. Consequently, any equal-length pair in distinct isometry orbits has intersection at least twenty-five.*

Proof. Distinct essential simple classes on a punctured torus have positive intersection. Normalize their vectors to $(1, 0)$ and (r, N) . The sign-and-inversion orbit (5) exhausts the permitted changes of marking of an unordered unoriented pair. Reflection residues are covered by Lemma 7.2. For every remaining orbit at $N \leq 24$, Table 2 gives its least positive representative.

Every listed representative $h \leq 5$ satisfies $N = hk + t$ with $k \geq 1$ and $0 < t < h$ coprime to h . The corresponding simple curve is the literal word (6), so Theorem 7.1 applies in the changed marking. Its actual isometry transports back to the original pair. Two representatives are left: 7 at $N = 19$, which is Proposition A.1, and 7 at $N = 23$, which is Proposition A.2. At $N = 24$ every unit squares to one modulo 24, so the reflection lemma alone settles that intersection. These cases exhaust all unit residues. At $N = 1$ the integral reflection formula applies directly.

The partition of the residues into orbits, the integral matrix realizing each sign or inversion change, and the mechanical gap pattern, remainder and quotient at every application of the occurrence theorem, are all finite assertions about integers below 25;

they were verified by computer in exact arithmetic. No Markov label is evaluated at this step. $\square$

N	Nonreflection representatives	N	Nonreflection representatives
1	none	12	none
2	none	13	2, 3, 5
3	none	14	3
4	none	15	2
5	2	16	3
6	none	17	2, 3, 4, 5
7	2	18	5
8	none	19	2, 3, 4, 7
9	2	20	3
10	3	21	2, 4
11	2, 3	22	3, 5
		23	2, 3, 4, 5, 7
		24	none

TABLE 2. The complete sign-and-inversion orbit cover through intersection twenty-four, after removing all residues with $r^2 \equiv 1 \pmod{N}$. Every representative at most five is covered by the occurrence theorem; the two representatives seven are the analytic intersection-nineteen and intersection-twenty-three cases. At twenty-four every unit is a reflection residue.

The first residue orbits beyond this cover are the two at intersection twenty-five,

$$\mathcal{O}_{25}(7) = \{7, 18\}, \quad \mathcal{O}_{25}(9) = \{9, 11, 14, 16\}.$$

The remaining nonreflection orbits modulo twenty-five have representatives 2, 3, 4 and are already covered. The first of the two is the pair $\{r, -r^{-1}\}$ of a residue with $r^2 \equiv -1$, which is why the sign-and-inversion orbit has two elements instead of four; the second has full size. The effective theorem supplies a finite decision for occurrence counts seven and nine, but the present classification has not executed those domains. An unclassified orbit is not evidence that a nonisometric equality exists there. The global Markov–Frobenius uniqueness problem remains open.

8. WHAT REMAINS AFTER EFFECTIVE FINITENESS

The results separate two questions that are often conflated. At each fixed occurrence count, equality is completely decidable by a finite calculation with an explicit polynomial bound. To settle Markov uniqueness, one must control the entire sequence of these finite questions as the count grows. Theorem 6.3 answers the first question and supplies a precise setting for the second.

The restriction is geometric, not just computational. The residue orbit (5) is invariant under a common change of marking of the pair. A change of basis can simplify the matrix character or the word, but cannot in general reduce the least representative of that orbit. Thus the method does not become a proof of uniqueness by repeatedly choosing a new convenient basis. The remaining intersection range is also unbounded: the result through 24 is a uniform theorem at those intersections, not an estimate on all possible intersections.

Example 8.1 (Isometric pairs at every occurrence count). For any $h \geq 2$, take root homology columns $v = (h+1, 1)$ and $b = (-1, 0)$. They form a simple marking since $\det(v, b) = 1$. The actual order-three action G from Proposition 6.4 satisfies

$$Gv = (-1, h) = hv + (h^2 + h + 1)b.$$

Thus an isometric equal-length pair has original type $(k, t) = (h+1, 1)$ at every occurrence count. For $N = h^2 + h + 1$, the sign-and-inversion orbit is $\{h, h+1, N-h, N-h-1\}$, whose least positive representative is h . These pairs therefore retain unbounded least occurrence count under common changes of marking.

8.1. An unbounded family beyond the finite cover. Proposition A.2 is the second member of a family, and the family is infinite. Keep $X = AB^3$ and $Y = AB^4$, and for $n \geq 0$ compare A with $X(X^nY)^2$. The pair (X, X^nY) is again a simple basis, so the second word is primitive and simple; its homology is $(2n+3, 6n+11)$, and its two long gaps are separated by n and $n+1$ short ones, so the cyclic gap pattern is mechanical. Both the occurrence count and the intersection number therefore grow without bound along the family. At $n = 0$ the pair is the three-occurrence type $(k, t) = (3, 2)$, settled by theorem 7.1; at $n = 1$ it is the residue-five pair at intersection seventeen, also settled there; at $n = 2$ it is proposition A.2.

We have proved that $\text{tr } A \neq \text{tr}(X(X^nY)^2)$ for every $n \geq 0$ and every positive integral marked character, so that the residue h at intersection $3h+2$ carries no equal-length pair for any odd $h \geq 3$. The argument beyond $n = 2$ is not the one used here: it replaces the fixed elimination by a spectral representation of the fixed-coordinate ray, with uniform bounds on its terminal core, and it is accompanied by an elementary congruence filter modulo nine which by itself already removes every quotient divisible by four and every quotient congruent to six modulo twelve. Because that machinery is of a different kind and about as long as the present article, it is written up separately. We record the statement here because it shows what the finite cover of theorem 7.3 does not see: an unbounded prescribed family can be excluded outright, while the cover itself stops at twenty-four.

This does not bound the occurrence count of an arbitrary equality, and it does not cover every residue at the intersections it meets.

The error estimates point to a more specific analytic question. An isometric equality can occur inside the finite domain and must be retained. Any uniform refinement must therefore distinguish the actual symmetry equations from a possible nonisometric equality. Excluding all equalities at large occurrence count would be too strong. A promising next step is to characterize equality in the dominant-state and Euclidean-descent constraints in terms of the twelve actions of Proposition 6.4. The present bounds establish neither that characterization nor a bound on the first possible nonisometric occurrence count.

There is a second reformulation of the same uniqueness question, arithmetic rather than geometric, and it is worth recording here because it shows from a different side why a bound on the occurrence count is the missing ingredient and not a technicality. A simple curve is presented by a Christoffel word, and Frobenius attached to the corresponding Markov number m a square root of -1 modulo m [7]. That root is not an abstract residue: it is a letter count. Writing a proper Christoffel word as $a\text{Pal}(d)b$ for a directive word d , with q letters a and s letters b , the word directed by the doubled directive $\hat{d}d$, where $\hat{d}$ is the reversal complement, has length exactly $m = q^2 + s^2$ and precisely sq^{-1} letters b ; and, the incidence matrix of a doubled directive being a Gram matrix and hence symmetric, that same number says where the word splits in its standard factorization into two Christoffel words. The classical dictionary behind these statements is the involution theory of Christoffel words and Sturmian morphisms [2, Section 3] together with Reutenauer's correspondence [13]; the proofs of the two statements just

quoted are given in a companion article of this series. A combinatorial proof that a Markov number is a sum of two squares was obtained independently by L. Vuillon and A. De Luca (personal communication, 2026).

Read through that dictionary, uniqueness becomes the assertion that a Markov number determines the primitive representation $m = q^2 + s^2$ of every occurrence carrying it. Since m has $2^{\omega(m)-1}$ primitive representations, with ω the number of distinct prime divisors, the question is empty when m is a prime power or twice one, which recovers the classical results in that range and explains why they stop there. What it does not supply is any control as $\omega(m)$ grows, and in particular no bound on the occurrence count. The reformulation and the present theorems therefore fail in the same place, from opposite directions: the arithmetic side decides a single label and cannot follow a family, while [theorem 6.3](#) decides a whole family at a fixed count and cannot follow the count.

The contribution of the method is that the trace comparison keeps three kinds of structure visible at once: the positive paths of the word, the integral Vieta orbit of its character, and the marking that recognizes a surface symmetry. The positive paths and integral Vieta orbit produce the terminal contradiction and hence the parameter bounds; the marking identifies the symmetries among the retained equalities. Together they reduce an unbounded search in matrix entries and gap lengths to an explicit family of finite geometric decisions, while leaving the remaining uniform question identifiable.

APPENDIX A. THE FINITE DOMAINS AND THE INTERSECTION-NINETEEN ARGUMENT

The finite part of the rigidity theorem has two different roles. For a fixed occurrence count, the analytic bounds reduce the problem to a finite list of minimum seeds and trace comparisons. At intersection nineteen, four polynomial sign conditions establish inequalities on entire unbounded integer cones. We describe the two separately, because a finite trace comparison never substitutes for an argument valid on an unbounded cone.

A.1. The finite occurrence domains. For a given occurrence count h , put

$$H_+(h) = 4^{h+1} - 1, \quad H_0(h) = (100h^2)^{h+1} - 1.$$

The parameter bounds in the main text give

$$l \geq 1: \quad p \leq H_+(h), \quad 1 \leq l \leq 3h - 1, \quad -\left\lceil \frac{h}{2} \right\rceil \leq q \leq \left\lfloor \frac{3h}{2} \right\rfloor, \quad (92)$$

$$l = 0: \quad p \leq H_0(h), \quad -\left\lceil \frac{h}{2} \right\rceil \leq q \leq \left\lfloor \frac{5h}{2} \right\rfloor. \quad (93)$$

For each $1 \leq t < h$ with $\gcd(h, t) = 1$, set $m = hl + t + q$, retain only $m \geq 1$, and compare both u_m and u_{-m} with the literal Christoffel product. Reversing the minimum ray changes the word as well as the single index. If that index is $n = \pm m$, the original quotient and remainder are

$$(k, t_{\text{orig}}) = \begin{cases} (l - n, t), & l - n \geq 1, \\ (n - l - 1, h - t), & n - l - 1 \geq 1. \end{cases} \quad (94)$$

These alternatives are disjoint; rows satisfying neither are outside the original positive-quotient problem.

[Lemma 6.1](#) provides a finite tree of occurrences, not a set of numerical labels. At a node (a, b, p) with a basis (U, V) having character $3(a, b, p)$, the two minimum-ray choices are $A = U$ and $B = (UV)^{\pm 1}$. Their initial values are $x = a$ and respectively $y = 3pa - b$ or $y = b$. Both satisfy $x \leq y$ and $x \leq 3px - y$. The traversal preserves distinct nodes with the same numerical maximum. Thus it never assumes the label uniqueness motivating the paper.

The traces are computed in two independent ways. The first multiplies the integral matrices of each seed. The second generates the Markov tree afresh and works in the character algebra with basis (I, X, Y, XY) . If $x_1 = \text{tr } X$, $y_1 = \text{tr } Y$, $z_1 = \text{tr}(XY)$, right multiplication acts by

$$(a, b, c, d)X = (-b + (z_1 - x_1 y_1)c - y_1 d, a + x_1 b + y_1 c + z_1 d, x_1 c + d, -c), \quad (95)$$

$$(a, b, c, d)Y = (-c, -d, a + y_1 c, b + y_1 d). \quad (96)$$

The trace is $2a + x_1 b + y_1 c + z_1 d$. Here

$$x_1 = 3u_l, \quad y_1 = 3u_{l+1}, \quad z_1 = 3(3u_l u_{l+1} - p).$$

The second calculation therefore shares neither the matrix arithmetic nor the seed traversal of the first. For every equality one records the full marked character and the two primitive homology vectors and exhibits one of the isometries of Proposition 6.4 carrying the first vector to the second; equality of numerical labels is not itself an isometry test. The equality records are in the supplementary material.

A.2. The remaining residue at intersection nineteen. The occurrence theorem covers every nonreflection residue at intersection at most twenty-four except one orbit at nineteen and one at twenty-three. The first is treated here and the second in A.3. Write $X = AB^2$, $Y = AB^3$. Representatives of this orbit are

$$U = (XY^{-1})^2 X, \quad V = (XY^2)^2 Y. \quad (97)$$

Their vectors in the (X, Y) basis are $(3, -2)$ and $(2, 5)$, with determinant nineteen. In the original basis they are $(1, 0)$ and $(7, 19)$.

Throughout this subsection the letters A, B, C, D name the coefficients of the eliminated quadratic, and R, T two auxiliary polynomials in the single variable s . None of them carries the meaning it has in the main text, where A, B are the two basis matrices and T the modular torus; the two settings never meet, because the argument that follows is purely polynomial. The shifted variables u, v, w are likewise local to the computation in which each appears.

Proposition A.1. *Let (X, Y) be a positively marked modular-torus basis, and let U, V be the primitive classes in (97). If $\text{tr } U = \text{tr } V$, then*

$$\frac{1}{3}(\text{tr } X, \text{tr } Y, \text{tr}(XY)) = (1, 1, 1).$$

Both traces are 87, and an order-three modular isometry carries one curve to the other.

Proof. Put $(r, s, t) = \frac{1}{3}(\text{tr } X, \text{tr } Y, \text{tr}(XY))$. These are positive integers with $r^2 + s^2 + t^2 = 3rst$. Define

$$P(r, s, t) = (9(3rs - t)^2 - 1)r - 3(3rs - t)s. \quad (98)$$

Cayley–Hamilton gives $m := \text{tr } U/3 = P(r, s, t)$ and $n := \text{tr } V/3 = P(s, t, r)$. We treat both orders of r, s .

The order $r \geq s$. Set $p = 3rs - t$. Eliminating p gives the necessary equation

$$Q(r, s) = 81r^5(r - R(s)) + 27r^4 + A(s)r^3 + B(s)r^2 - C(s)r + D(s) = 0, \quad (99)$$

where

$$\begin{aligned} R(s) &= 729s^7 - 567s^5 + 126s^3 - 7s, \\ A(s) &= 32805s^7 - 13851s^5 + 1134s^3 + 9s, \\ B(s) &= 324s^4 + 1, \\ C(s) &= 3645s^7 - 567s^5 + 27s^3 + 2s, \\ D(s) &= 6561s^{10} - 1458s^8 + 243s^6 - 9s^4 + s^2. \end{aligned}$$

To include degenerate elimination coefficients, put $f = p^2 + r^2 + s^2 - 3prs$. Polynomial division gives $m - n = Hf + Lp + N_0$, with

$$\begin{aligned} H &= 9(r + s - 9s^3), \\ L &= 27r^2s + 243rs^4 - 81rs^2 + 3r - 3s, \\ N_0 &= s - r - 9r^2s(9s^2 - 1)(9s^2 - 2) - H(r^2 + s^2). \end{aligned}$$

The identity $Q = N_0^2 + 3rsLN_0 + (r^2 + s^2)L^2$ proves necessity even at $L = 0$; there has been no division by L .

Write $u = r - s$. Direct expansion gives

$$Q(s + u, s) = 81u^6 - \sum_{j=1}^5 d_j(s)u^j - 729s^6(s^2 - 1)(9s^2 - 2)^2, \quad (100)$$

where all five d_j are positive for $s \geq 1$. This is a finite assertion about integer coefficients: each $d_j(1 + v)$ has nonnegative coefficients and a positive constant. The coefficient lists are in the supplementary material, where they are regenerated from the literal words rather than read. Descartes' rule gives exactly one positive root in u . For $s = 1$ one first removes the factor u , after which there is again one sign change.

Let $w = R(s) - 1$. The following inequalities hold for every $s \geq 1$:

$$w > s, \quad C > 0, \quad A > C, \quad B > 0, \quad D > 0, \quad A < 81w, \quad B < 2w, \quad D < w^2. \quad (101)$$

Indeed each of

$$C, \quad A - C, \quad B, \quad D, \quad R - s - 1, \quad 81(R - 1) - A, \quad 2(R - 1) - B, \quad (R - 1)^2 - D$$

has nonnegative coefficients after $s = 1 + v$, with a positive constant. These eight full expansions accompany the five preceding ones. Also $w \geq 280$, since, for $v = s^2 - 1$,

$$R(s) - 281s^7 = sv(336 + 777v + 448v^2) \geq 0.$$

We have $Q(R(s), s) > 0$, because $AR^3 > CR$. At the other endpoint,

$$Q(w, s) < -81w^5 + 108w^4 + 2w^3 + w^2 \leq -\frac{211}{4}w^4 < 0;$$

the second inequality already holds for $w \geq 2$. The unique root with $r > s$ lies strictly between $R(s) - 1$ and $R(s)$, and cannot be integral.

On the diagonal, $Q(s, s) = -729s^6(s^2 - 1)(9s^2 - 2)^2$, so $r = s = 1$. The remaining equation gives $p = 1$ or $p = 2$. Their trace-label pairs are respectively $(m, n) = (5, 194)$ and $(29, 29)$. Thus only $p = 2$, or $t = 1$, is an equality. In particular, a root of the eliminated polynomial alone is insufficient.

The order $s > r$. We preserve the actual Markov sheet by positive Vieta mutations. Recall that for an ordered positive integral Markov triple $a \leq b < c$, its replacement $3ab - c$ is positive and at most b ; equality occurs only at the initial repeated-entry triples. To see the upper bound, evaluate $z^2 - 3abz + a^2 + b^2$ at $z = b$; the value is $a^2 - (3a - 2)b^2 \leq 0$. Positivity follows from $c(3ab - c) = a^2 + b^2$.

Keep $p = 3rs - t$ and put $z = 3st - r$. If $t \geq r$, then

$$z - p = (3s + 1)(t - r) \geq 0, \quad z \geq (3s - 1)t.$$

Since $s \geq 2$ and $s - r \geq 1$,

$$\begin{aligned} n - m &\geq (9z^2 - 1)(s - r) - 3zt + 3ps \\ &> 9z^2 - 1 - \frac{3z^2}{3s - 1} \geq \frac{42}{5}z^2 - 1 > 0. \end{aligned}$$

Equality therefore forces $t < r < s$, and $r \geq 2$. Make two mutations

$$w = 3rt - s, \quad v = 3tw - r.$$

Descent gives $w < r$ and $v \leq \max(t, w)$. If $w \geq t$, then

$$r = 3tw - v \geq (3t - 1)w \geq (3t - 1)t.$$

Also $p > (3s - 1)r$, whence $pr > 4(3s - 1) > 3s + 1$ and $r/p < 1$. These imply $p^2r > 3ps + r$, and consequently $m > 8p^2r$, whereas $n < 9z^2s$. Using

$$tp - rz = r^2 - t^2 > 0, \quad s < 3tr,$$

we obtain

$$\frac{m}{n} > \frac{8}{9} \left(\frac{p}{z} \right)^2 \frac{r}{s} > \frac{8r^2}{27t^3} \geq \frac{8(3t - 1)^2}{27t} \geq \frac{32}{27} > 1.$$

The last step uses $(3t - 1)^2 - 4t = (t - 1)(9t - 1) \geq 0$. This contradiction proves $t > w$. Descent now also gives $v < t$, so the next mutation

$$a = 3wv - t$$

satisfies $1 \leq a \leq \max(w, v)$.

Define the single polynomial

$$\begin{aligned} T &= 3wv - a, & R_1 &= 3Tw - v, & S_1 &= 3TR_1 - w, \\ \mathcal{D}(a, w, v) &= P(R_1, S_1, T) - P(S_1, T, R_1). \end{aligned} \tag{102}$$

At our actual triple $(R_1, S_1, T) = (r, s, t)$, so $\mathcal{D} = m - n$ exactly. Its total degree is nineteen and it has 151 nonzero terms. No factor or denominator has been removed.

For distinct w, v , Table 3 covers all positive integers a, w, v with $a \leq \max(w, v)$. Each parameter A, B, C is a nonnegative integer. Expanding (102) under these four substitutions gives the indicated strict coefficient signs, including a nonzero constant of the same sign. Thus the polynomial has that strict sign throughout the entire cone, which contradicts $m = n$.

Order	(a, w, v)	Sign	Terms
$a \leq w < v$	$(A + 1, A + B + 1, A + B + C + 2)$	negative	1234
$w < a \leq v$	$(A + B + 2, A + 1, A + B + C + 2)$	negative	659
$a \leq v < w$	$(A + 1, A + B + C + 2, A + B + 1)$	positive	1436
$v < a \leq w$	$(A + B + 2, A + B + C + 2, A + 1)$	positive	1091

TABLE 3. The four descended integer cones. For $A, B, C \geq 0$, every nonzero coefficient, including the constant, has the listed sign. All 4420 coefficients are supplied as exact integers.

It remains to consider $v = w$. A repeated entry in a positive integral Markov triple equals one: $t^2 = w^2(3t - 2)$ gives $w \mid t$ by prime valuations; writing $t = bw$ gives

$b^2 + 2 = 3bw$, so $b \mid 2$, and substitution yields $w = 1$ and $t \in \{1, 2\}$. Since $t > w$, only $t = 2$ remains. Inverting the literal mutations gives $(r, s, t) = (5, 29, 2)$, but

$$P(5, 29, 2) - P(29, 2, 5) = 945951 \neq 0.$$

There is no equality in the second ordering.

The surviving equality is isometric. At $(r, s, t) = (1, 1, 1)$ take

$$E = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad F = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}, \quad K_0 = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix},$$

in the notation of the proof of Proposition 6.4. They satisfy $K_0 E K_0^{-1} = F$, $K_0 F K_0^{-1} = (EF)^{-1}$ and $K_0^3 = -I$, so K_0 is an actual modular-torus normalizer. Its homology action is the matrix G of (89), and

$$G = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix} \quad \text{satisfies} \quad G \begin{pmatrix} 3 \\ -2 \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}.$$

It maps the two primitive classes in (97) to one another. Marked-character conjugacy transports this isometry to the original basis. Direct traces give 87 for both words and the original character $3(29, 2, 5)$. $\square$

A.3. The remaining residue at intersection twenty-three. Write $X = AB^3$, $Y = AB^4$. The mechanical pattern of homology $(7, 23)$ is the cyclic class of $X^3 Y X^2 Y$: its seven gaps are five threes and two fours, with the two long gaps as evenly separated as the pattern allows. The pair $(X, X^2 Y)$ is again a simple basis, and in it that word is $X(X^2 Y)^2$, so the literal word, and not merely its abelianization, represents a primitive simple class. The two classes compared are therefore

$$U = A, \quad V = X^3 Y X^2 Y = X(X^2 Y)^2, \quad (103)$$

with homology vectors $(1, 0)$ and $(7, 23)$ in the original basis. Note also that $XY^{-1} = ABA^{-1}$ up to inversion, so (r, s, p) below is again a positive integral Markov triple, now for the basis (X, Y) .

As in A.2, the letters A, B and the auxiliary polynomials below are local to this subsection.

Proposition A.2. *Let (A, B) be a positively marked modular-torus basis and let U, V be the primitive classes in (103). Then $\text{tr} U \neq \text{tr} V$ in every positive integral marked character. Consequently the orbit $\{7, 10, 13, 16\}$ modulo twenty-three carries no equal-length pair.*

Proof. Put $p = \frac{1}{3} \text{tr} B$, $r = \frac{1}{3} \text{tr} X$ and $s = \frac{1}{3} \text{tr} Y$. These are positive integers with $p^2 + r^2 + s^2 = 3prs$. The trace recurrence for $X^j Y$ and the identity $\text{tr}(XW^2) = \text{tr} W \text{tr}(XW) - \text{tr} X$ with $W = X^2 Y$ give

$$m := \frac{1}{3} \text{tr} U = (27p^3 - 6p)r - (9p^2 - 1)s, \quad (104)$$

and, with $a = \frac{1}{3} \text{tr}(X^2 Y)$ and $b = \frac{1}{3} \text{tr}(X^3 Y)$, that is

$$a = (9r^2 - 1)s - 3pr, \quad b = (27r^3 - 6r)s - (9r^2 - 1)p,$$

the competing label $n := \frac{1}{3} \text{tr} V = 3ab - r$. Positive integral Vieta dynamics keeps all of p, r, s, a, b positive, and equality of lengths means $m = n$.

Reduce $F = m - n$ modulo $\mathcal{M} = p^2 + r^2 + s^2 - 3prs$, viewed as a monic quadratic in p . The remainder is $Lp - N$ with

$$\begin{aligned} L &= 3(81r^4s + 81r^3s^2 - 9r^3 - 27r^2s - 18rs^2 - 2r + s), \\ N &= 729r^5s^2 - 81r^5 + 81r^4s - 324r^3s^2 + 9r^3 + 81r^2s^3 \\ &\quad - 9r^2s + 27rs^2 - r - 9s^3 - s. \end{aligned}$$

Multiplying $\mathcal{M} = 0$ by L^2 therefore turns every equality into

$$Q(r, s) = N^2 - 3rsNL + (r^2 + s^2)L^2 = 0. \quad (105)$$

No division has been performed, so (105) remains necessary when $L = 0$. We prove the stronger statement that Q has no positive integral zero at all, with no Markov hypothesis on (r, s) .

One real root in each ordering. For real $a \geq 1$ write

$$Q(a, a+u) = 81u^6 - \sum_{j=0}^5 b_j(a)u^j, \quad Q(a+u, a) = 6561u^{10} - \sum_{j=0}^9 c_j(a)u^j.$$

Every b_j and every c_j is positive on $a \geq 1$: after removing the factor $a^{j \bmod 2}$ and substituting $z = a^2 - 1$, each becomes a polynomial in z with positive coefficients, and the fifteen coefficient lists are in the supplementary material. The constant terms are therefore negative and the leading terms positive, so Descartes' rule gives exactly one positive root of each shifted polynomial. On the diagonal $Q(a, a) < 0$.

The root with larger second coordinate. Let

$$R = 6561r^9 - 6561r^7 + 2187r^5 - 270r^3 + 9r,$$

which is $\frac{1}{3} \operatorname{tr}(C^9)$ whenever $\det C = 1$ and $\operatorname{tr} C = 3r$. The identity

$$R - 1926r^9 = r(r^2 - 1)(4635r^6 - 1926r^4 + 261r^2 - 9)$$

has a positive last factor for $r \geq 1$, because $4635r^6 - 1926r^4 \geq 2709r^4$ and $261r^2 > 9$; hence $R - 1 \geq 1925r^9 > r$. Grouping,

$$Q = 81s^5(s - R) + A_4s^4 + A_3s^3 + A_2s^2 + A_1s + A_0$$

with $A_4 = 324r^2 + 27$, $A_2 = 972r^4 + 1$, $A_0 = 6561r^{10} - 729r^8 + 567r^6 + 18r^4 + r^2$,

$$A_3 = 295245r^9 - 164025r^7 + 19683r^5 + 486r^3 - 63r,$$

$$A_1 = -32805r^9 + 3645r^7 + 729r^5 - 54r^3 + 2r.$$

Here $A_0, A_2, A_4 > 0$, while $A_3 \geq 131157r^9$ and $A_1 \geq -32859r^9$, so $A_3s^3 + A_1s > 0$ for $s \geq 1$ and $Q(r, R) > 0$. At $w = R - 1 \geq 1925r^9$, discarding every negative contribution in the remainder and dividing by w^5 ,

$$\frac{Q(r, w)}{w^5} \leq -81 + \frac{351}{1925} + \frac{315414}{1925^2} + \frac{973}{1925^3} + \frac{4376}{1925^4} + \frac{7147}{1925^5} < -80 < 0,$$

each term obtained by replacing lower powers of $r \geq 1$ by the largest numerator power and using $w \geq 1925r^9$. The positive sum is exactly $1009966664583171/3776205576171875 < 1$. The unique root above $s = r$ thus lies strictly between $R - 1$ and R , so no integral s meets it when r is an integer.

The root with larger first coordinate. Let $T = 81s^5 - 45s^3 + 5s$, which is $\frac{1}{3} \operatorname{tr}(C^5)$ at $\det C = 1$, $\operatorname{tr} C = 3s$. From $T - 41s^5 = 5s(s^2 - 1)(8s^2 - 1)$ we get $T - 1 \geq 40s^5 > s$. Grouping the same polynomial in r ,

$$\begin{aligned} Q &= 6561r^9(r - T) - 729r^8 + B_7r^7 + 567r^6 + B_5r^5 \\ &\quad + (972s^2 + 18)r^4 + B_3r^3 + (324s^4 + 1)r^2 + B_1r + B_0, \end{aligned}$$

with

$$\begin{aligned} B_7 &= 531441s^5 - 164025s^3 + 3645s, & B_5 &= -177147s^5 + 19683s^3 + 729s, \\ B_3 &= 21870s^5 + 486s^3 - 54s, & B_1 &= -729s^5 - 63s^3 + 2s, \end{aligned}$$

and $B_0 = 81s^6 + 27s^4 + s^2$. At $r = T$ the first grouped pair is $-729r^8 + B_7r^7 = r^7(472392s^5 - 131220s^3) \geq 341172s^5r^7$, which exceeds the largest possible negative contribution $-177147s^5r^5$ of B_5r^5 ; and $B_3 \geq 21816s^5$ with $B_1 \geq -792s^5$ give $B_3r^3 + B_1r > 0$. Every other term is positive, so $Q(T, s) > 0$. At $w = T - 1 \geq 40s^5$, discarding negative terms and dividing by w^9 ,

$$\frac{Q(w, s)}{w^9} \leq -6561 + \frac{535086}{40^2} + \frac{567}{40^3} + \frac{20412}{40^4} + \frac{990}{40^5} + \frac{22356}{40^6} + \frac{325}{40^7} + \frac{2}{40^8} + \frac{109}{40^9} < -6226 < 0,$$

the positive sum being $87672906826504189/262144000000000 < 335$. The unique root above $r = s$ lies between $T - 1$ and T .

Neither ordering and neither diagonal admits a positive integral zero of Q , so $m = n$ is impossible. $\square$

The two analytic propositions of this appendix differ in their conclusions, and the difference is worth stating. At intersection nineteen one equality survives and is isometric; at intersection twenty-three none survives at all. Together with the occurrence theorem and the reflection lemma they close the cover at twenty-four, because every unit modulo twenty-four is a reflection residue.

A.4. What the polynomial computation establishes. The computation supporting Proposition A.1 reconstructs both literal traces from (95)–(96), with inverse letters handled by $X^{-1} = (\text{tr } X)I - X$, and then confirms the elimination identity, the five Descartes coefficient signs, the eight endpoint margins and both diagonal roots. The computation supporting Proposition A.2 is of the same kind and no larger: it confirms the reduction of F modulo $\mathcal{M}$, the necessary equation (105), the fifteen positive coefficient lists in $z = a^2 - 1$, the two power-trace identities for R and T , the four grouped sign estimates, and the two exact rational margins. The degree-nineteen polynomial (102) and its four substitutions are expanded separately, in sparse integer arithmetic; every exponent triple and coefficient is regenerated before comparison. The repeated-entry endpoint and the normalizer identities are confirmed in the same way.

These finite computations establish positivity only on the stated cones. The proofs above establish why every hypothetical equality reaches one of those cones and why the surviving homology action is an isometry. In particular, no census of Markov triples and no numerical approximation of a root enters either proposition: both roots are located between two consecutive integers by exact inequalities.

SUPPLEMENTARY MATERIAL

The supplementary material accompanying this article contains the exact computations used in the proofs of theorems 7.1 and 7.3 and propositions A.1 and A.2: the equality records with their isometry witnesses, the residue cover, the polynomial coefficient lists, and instructions for repeating the verification.

CODE AND DATA AVAILABILITY

The source code and the data supporting this article, with the instructions that repeat every calculation, are contained in the supplementary material of this article, which is available, together with that of the other articles of this series, at

<https://github.com/dutykh/fibonacci-pell-nearest-gaps/>

CONFLICT OF INTEREST

The authors declare that they have no conflict of interest.

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